Control of a Chemical Plant

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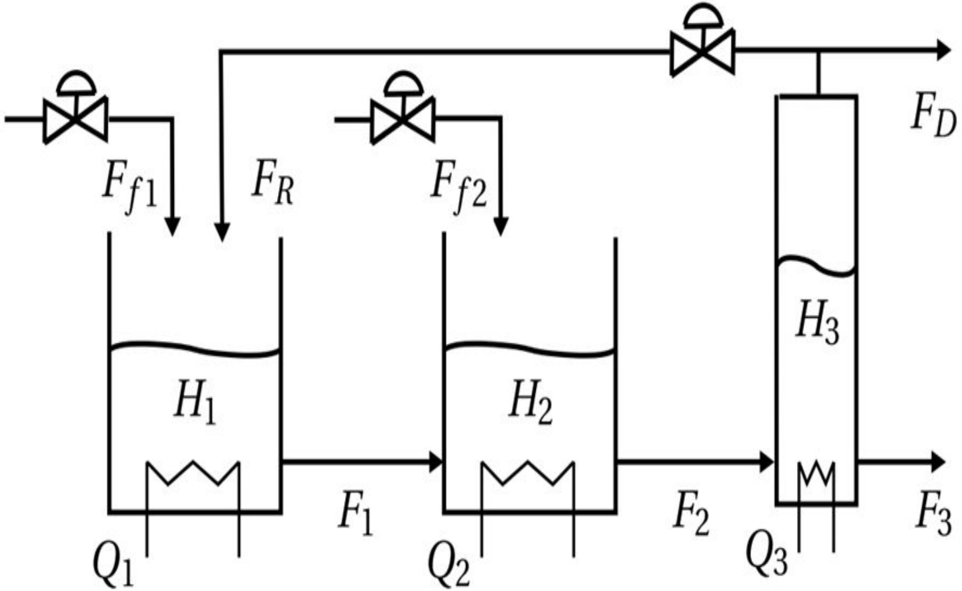
System Overview

System Configuration: The plant consists of three main interconnected unit operations: two chemical reactors arranged in series followed by a final separator.

In the analysis we considered the model of the system **linearized around the equilibrium position.**

Control Objective: The primary goal is to manage the output levels of reactors and seperator and product weight percent Xb3 by adjusting fresh feed flows Ff1, Ff2 recycle flow Fr and heat inputs Q1,Q2,Q3



$$y = [H_1 \quad x_{A1} \quad x_{B1} \quad T_1 \quad H_2 \quad x_{A2} \quad x_{B2} \quad T_2 \quad H_3 \quad x_{A3} \quad x_{B3} \quad T_3]$$

$$u = [F_{f1} \quad Q_1 \quad F_{f2} \quad Q_2 \quad F_R \quad Q_3].$$

Steady States Values

Steady states and parameters.

Parameter	Value	Units
H_1	29.8	m
x_{A1}	0.542	wt(%)
x_{B1}	0.393	wt(%)
T_1	315	K
H_2	30	m
x_{A2}	0.503	wt(%)
x_{B2}	0.421	wt(%)
T_2	315	K
H_3	3.27	m
x_{A3}	0.238	wt(%)
x_{B3}	0.570	wt(%)
T_3	315	K

Input constraints.

Parameter	Lower bound	Steady state	Upper bound	Units
F_{f1}	0	8.33	10	kg/s
Q_1	0	10	50	kJ/s
F_{f2}	0	0.5	10	kg/s
Q_2	0	10	50	kJ/s
F_R	0	66.2	75	kg/s
Q_3	0	10	50	kJ/s

Matrix Decomposition

Modeling: The plant is modeled as a large-scale system that can be partitioned into three independent subsystems (Reactor 1, Reactor 2, and the Separator) for distributed control analysis.

$$y_1 = [H_1 \quad x_{A1} \quad x_{B1} \quad T_1] \qquad u_1 = [F_{f1} \quad Q_1]$$
$$y_2 = [H_2 \quad x_{A2} \quad x_{B2} \quad T_2] \qquad u_2 = [F_{f2} \quad Q_2]$$
$$y_3 = [H_3 \quad x_{A3} \quad x_{B3} \quad T_3] \qquad u_3 = [F_R \quad Q_3].$$

Reactor 1
Reactor 2
Seperator

B =

2.2222	0	0	0	2.2222	0
0.0745	0	0	0	0.0734	0
0	0	0	0	0.0011	0
23.3314	0.0030	0	0	23.3314	0
0	0	2.2222	0	0	0
0	0	0.0740	0	0	0
0	0	0	0	0	0
0	0	23.1759	0.0030	0	0
0	0	0	0	-6.7333	0
0	0	0	0	-2.0304	0
0	0	0	0	-0.0303	0
0	0	0	0	-645.0579	0.0816

C =

1	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	0	0	0
0	0	0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0	0	1

Continuous-Time to Discrete-Time

$$F = e^{Ah}$$

$$G = \int_0^h e^{A\tau} d\tau \, B$$

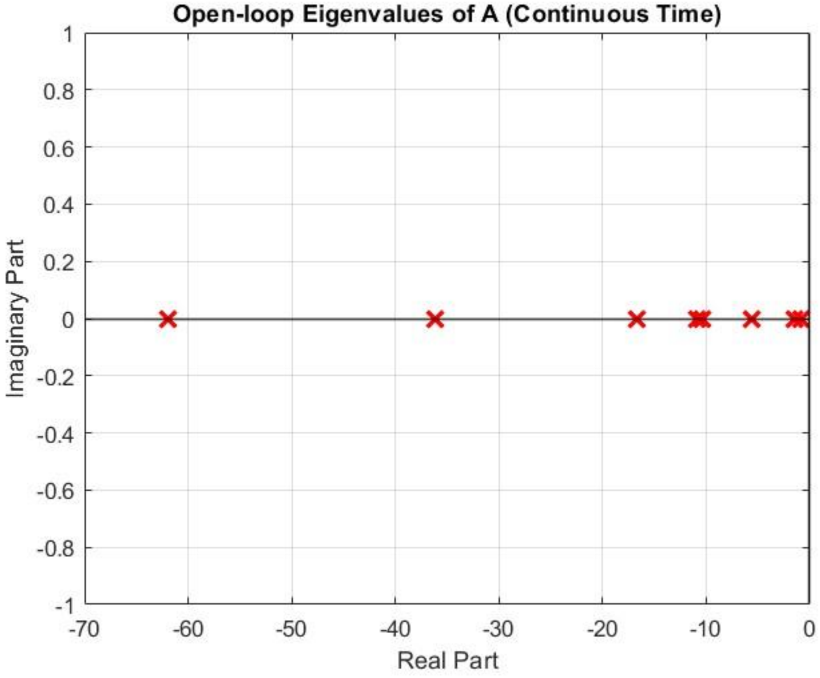
$$H = C$$

$$x_{k+1} = Fx_k + Gu_k$$
$$y_k = Hx_k$$

Zero-Order Hold Digital-Analog Converter
Sampling Time = 0.1 [s]

$$u(t) = u_k$$
$$t \in [kh, (k+1)h], \qquad where\ h = 0.1[s]$$

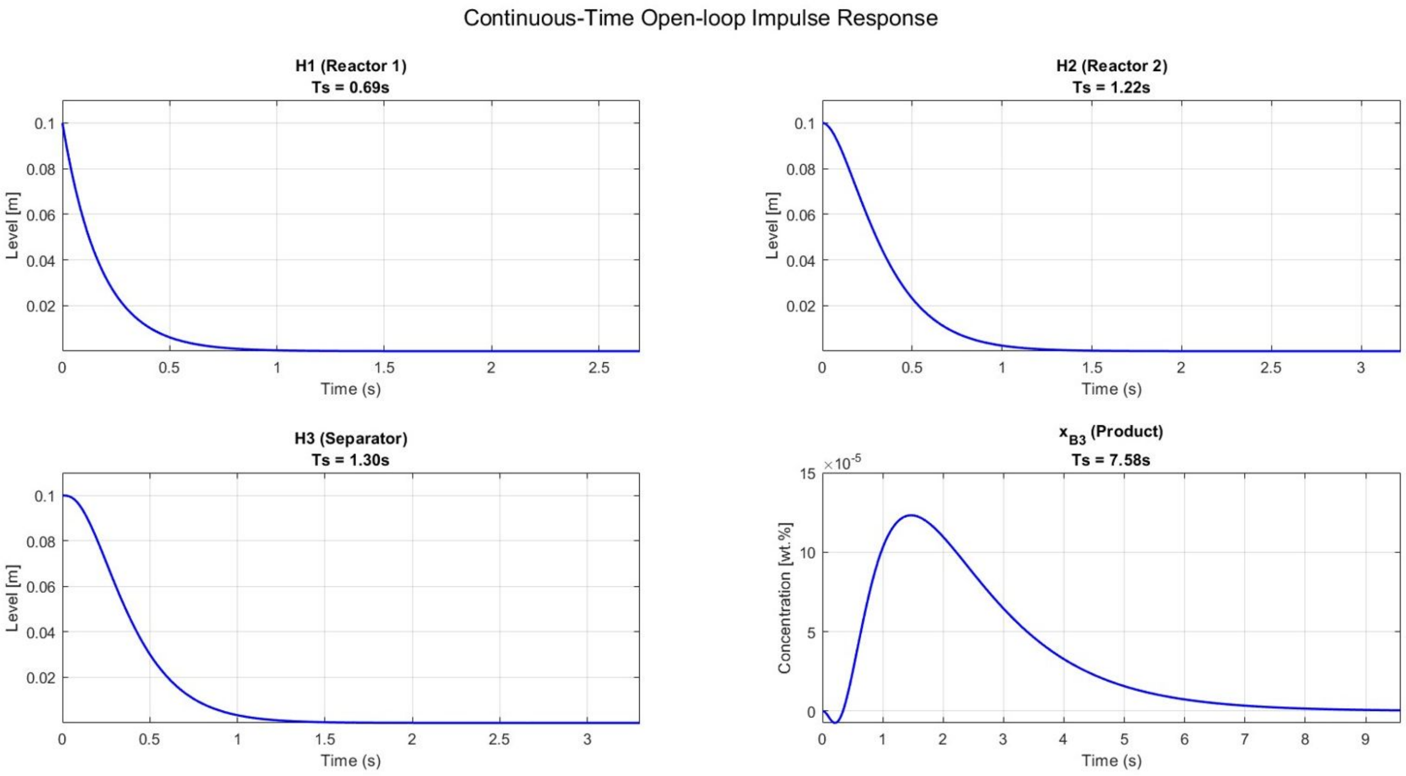
Continuous-Time Open-Loop Analysis



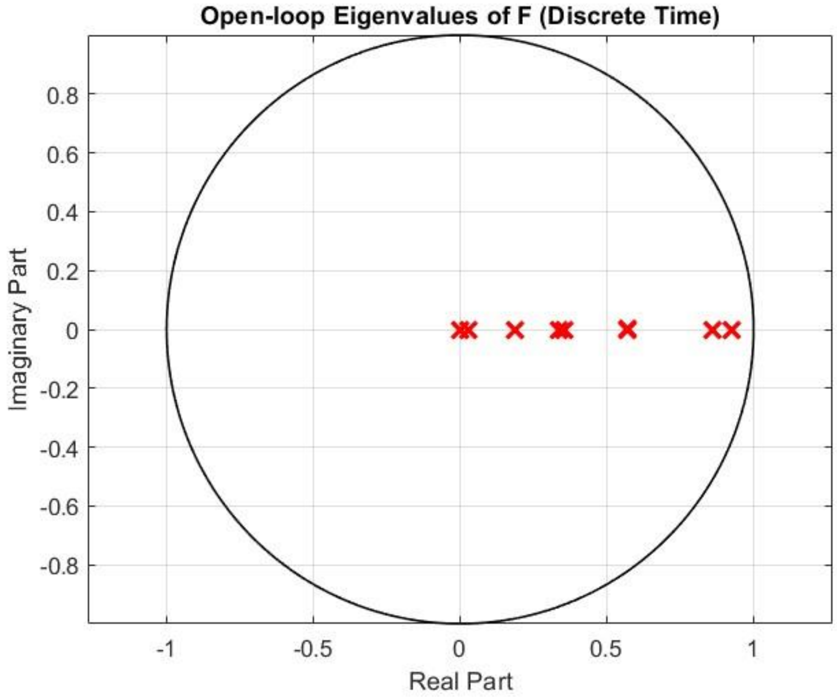

```
Continuous-time eigenvalues:
  -61.9492
  -36.1692
   -0.7802
   -1.5129
  -10.8707
  -10.3178
  -16.7000
   -5.6000
   -5.6000
  -16.7000
   -5.6000
   -5.6000

Spectral abscissa = -0.78015
System is CT asymptotically stable.
```

Perturbed System Response



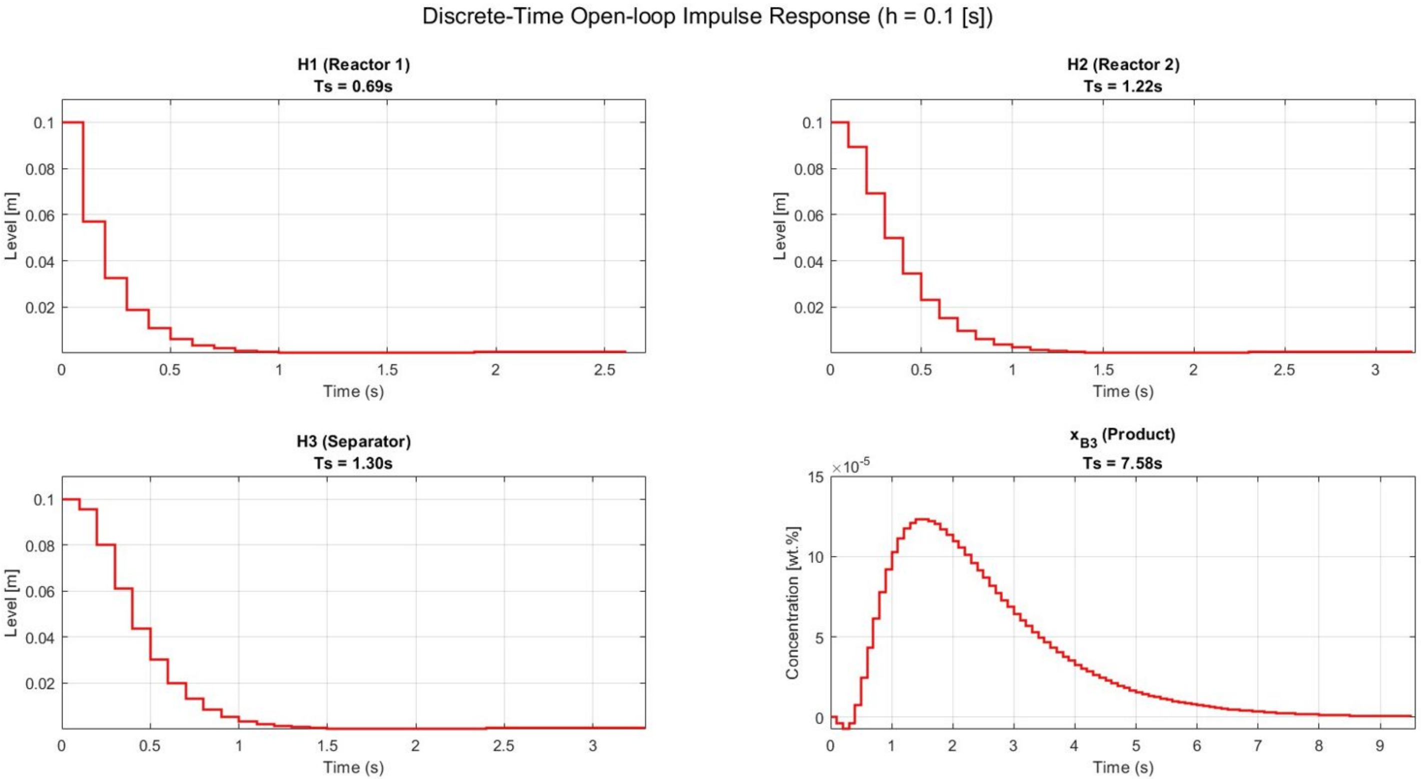
Discrete-Time Open-Loop Analysis




```
Discrete-time eigenvalues:
   0.9250 + 0.0000i
   0.8596 + 0.0000i
   0.1882 + 0.0000i
   0.1882 + 0.0000i
   0.0269 + 0.0000i
   0.0020 + 0.0000i
   0.3564 + 0.0000i
   0.3372 + 0.0000i
   0.5712 + 0.0000i
   0.5712 + 0.0000i
   0.5712 - 0.0000i
   0.5712 + 0.0000i

Spectral radius = 0.92495
System is DT asymptotically stable.
```

Perturbed System Response



Fixed Modes and Control Structure

```
A =

   1.0e+03 *

  -0.0056         0         0         0         0         0         0         0         0         0         0         0
  -0.0002   -0.0056         0         0         0         0         0         0         0    0.0008   -0.0009         0
        0         0   -0.0056         0         0         0         0         0         0   -0.0001    0.0016         0
  -0.0583         0         0   -0.0056         0         0         0         0         0         0         0         0
   0.0056         0         0         0   -0.0056         0         0         0         0         0         0         0
   0.0002    0.0055         0         0   -0.0002   -0.0056         0         0         0         0         0         0
        0         0    0.0055         0         0         0   -0.0056         0         0         0         0         0
   0.0579         0         0    0.0055   -0.0579         0         0   -0.0056         0         0         0         0
        0         0         0         0    0.0167         0         0         0   -0.0167         0         0         0
        0         0         0         0    0.0050    0.1531         0         0   -0.0049   -0.0386    0.0238         0
        0         0         0         0    0.0001         0    0.1531         0   -0.0002    0.0018   -0.0606         0
        0         0         0         0    1.5968         0         0    0.1531   -1.5978         0         0   -0.0167
```

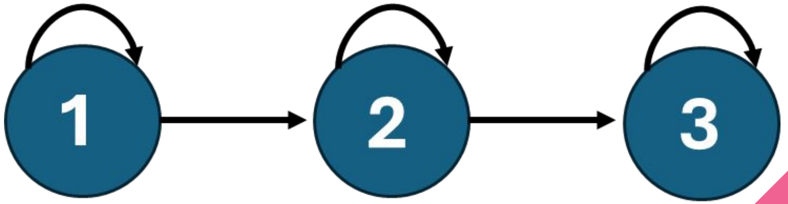
Centralized CT fixed modes:{ }

Centralized DT fixed modes:{ }

Decentralized CT fixed modes:{ }

Decentralized DT fixed modes:{ }

Distributed (“String”) CT and DT fixed modes:{ }

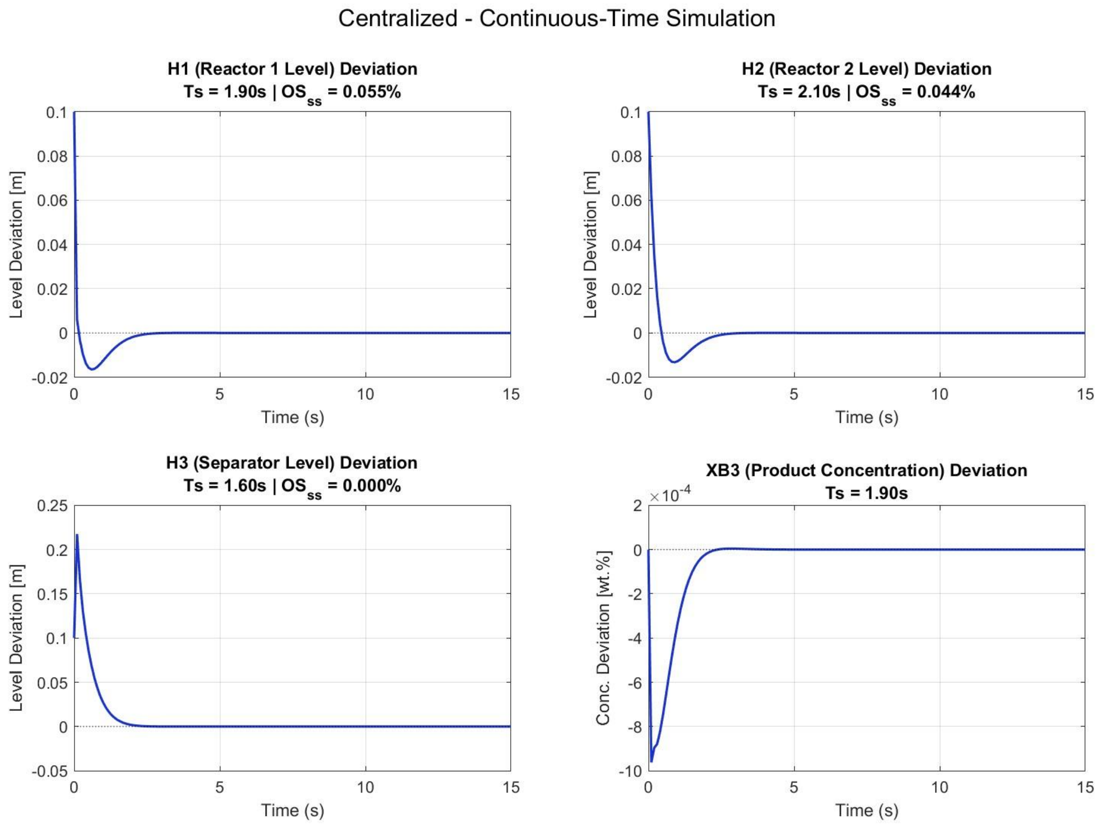


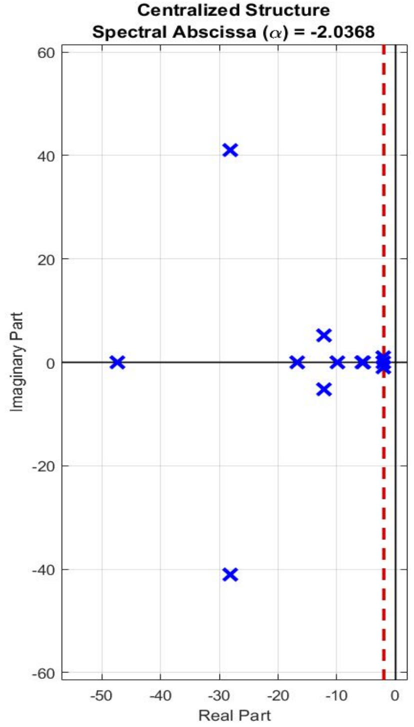
Settling Time	Overshoot	Control Effort	Input Constraints
$T_s \cong \frac{5}{Re(\lambda)}$ $T_s \leq 2.5\ s \qquad \alpha = 2$	$\zeta \in [\cos(\theta), 1]$ $\zeta = 0.707, where\ \theta = 45$ $OS\% = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$ x 100	$J = \alpha_L K_L + \alpha_Y K_Y$ $\alpha_L = 0.1, \alpha_Y = 10$	$\mu = \min(u_{\max} - u_{ss}, u_{ss} - u_{min})$ $(K_i x_i)^2 \leq \mu_i^2, where\ L = KY$ $\mu_i^2 - L_i Y^{-1} L_i^T \geq 0$
• $AY + YA + L^T B^T + BL + 2\alpha Y < 0$ • $Y > 0$	$\begin{bmatrix} \sin(\theta)\{AY + YA^T + BL + L^T B^T\} & \cos(\theta)\{AY - YA^T + BL - L^T B^T\} \\ \cos(\theta)\{-AY + YA^T - BL + L^T B^T\} & \sin(\theta)\{AY + YA^T + BL + L^T B^T\} \end{bmatrix} < 0$	• $\begin{bmatrix} K_L I & L^T \\ L & I \end{bmatrix} \geq 0$ • $\begin{bmatrix} K_Y I & I \\ I & Y \end{bmatrix} \geq 0$	• $\begin{bmatrix} \mu^2 & L \\ L^T & Y \end{bmatrix} \geq 0$

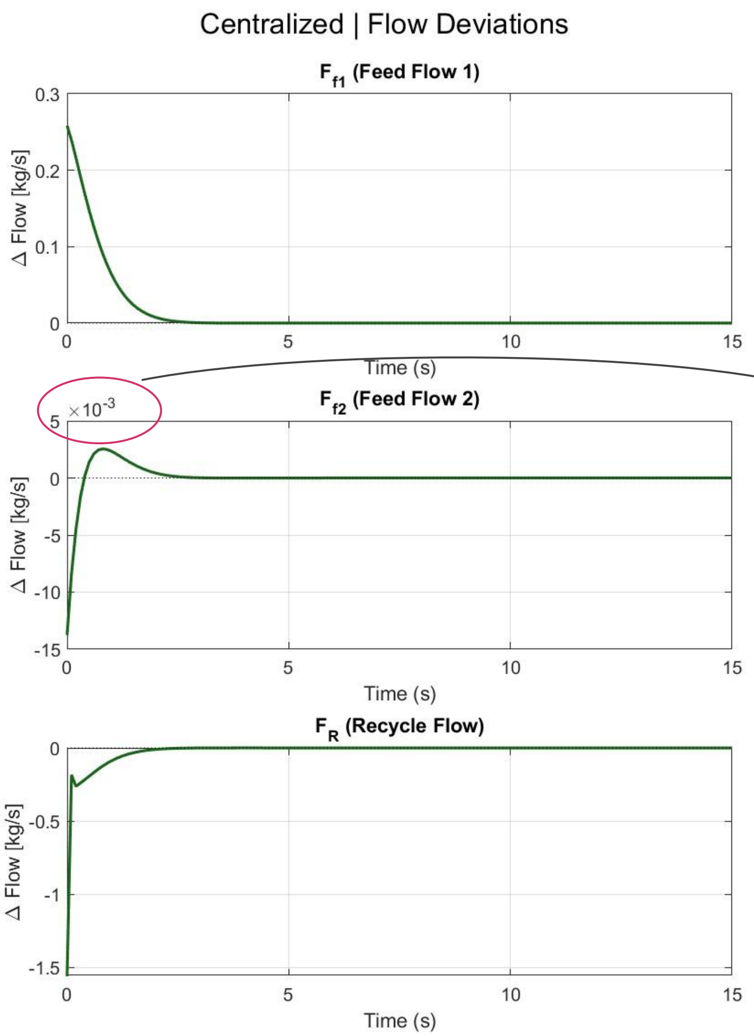
Centralized Control Fast Response without Damping

ℹ️Tradeoff: slow deviation
<-> well concentration

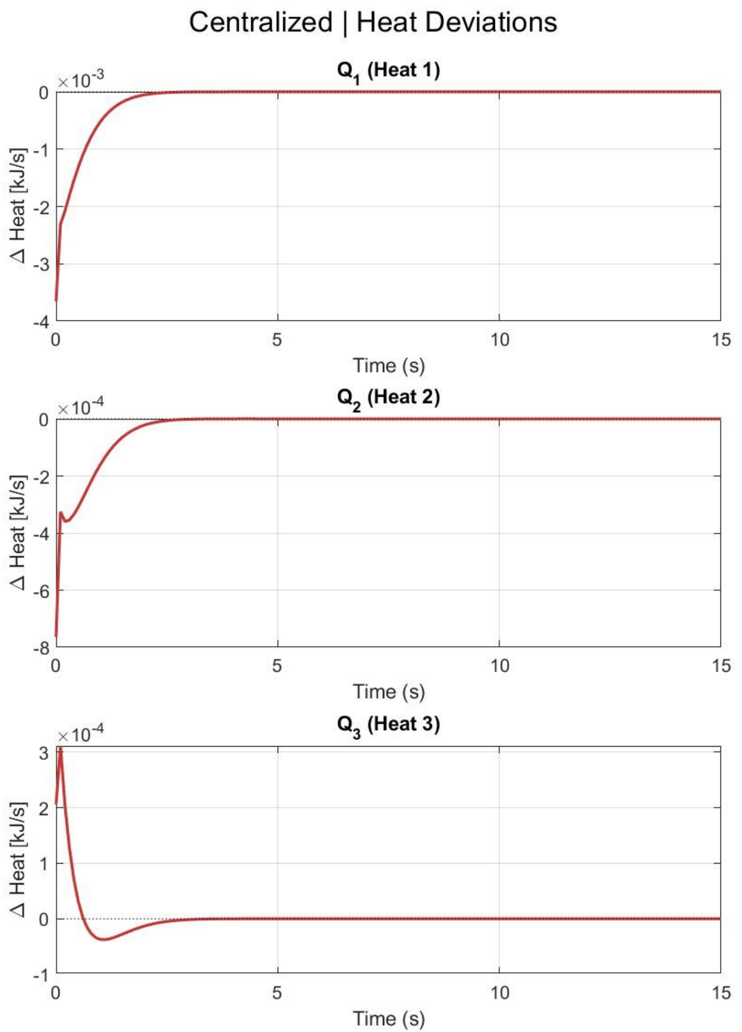
✅imaginary axis has ↑ contribution!



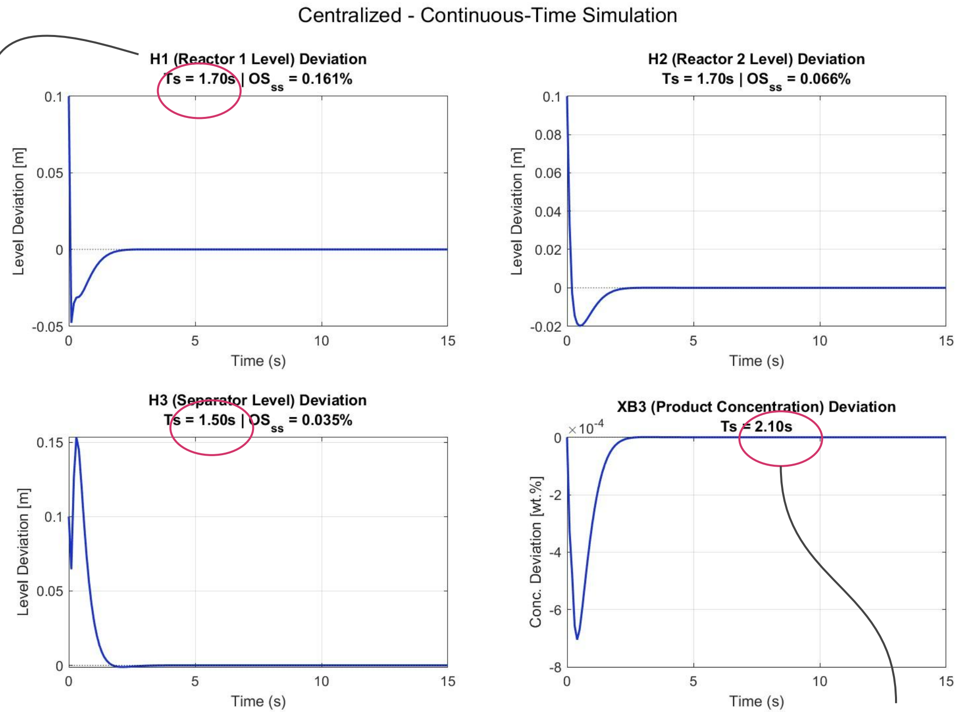




Inputs are usually between the boundaries (Ff2 has small deviation!)



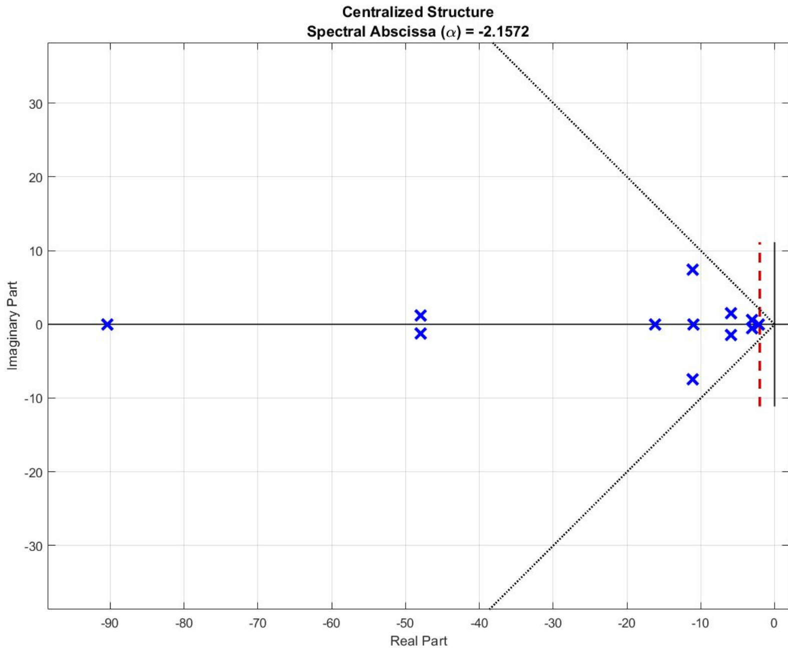
Continuous-Time , Fast Response with Damping

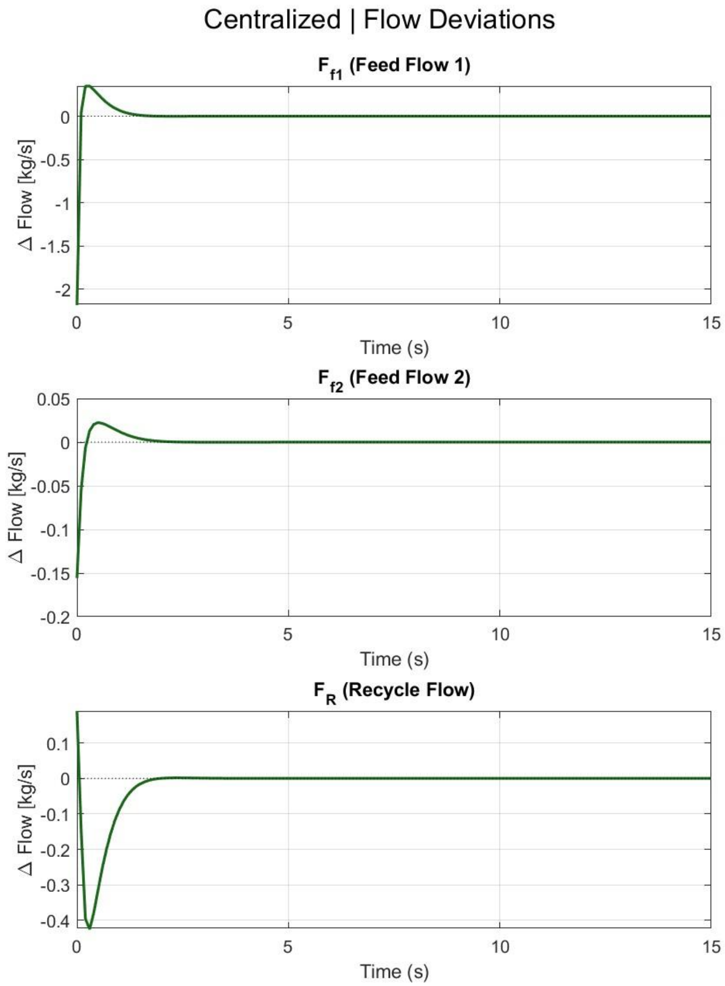


✅improvement

more aggressive

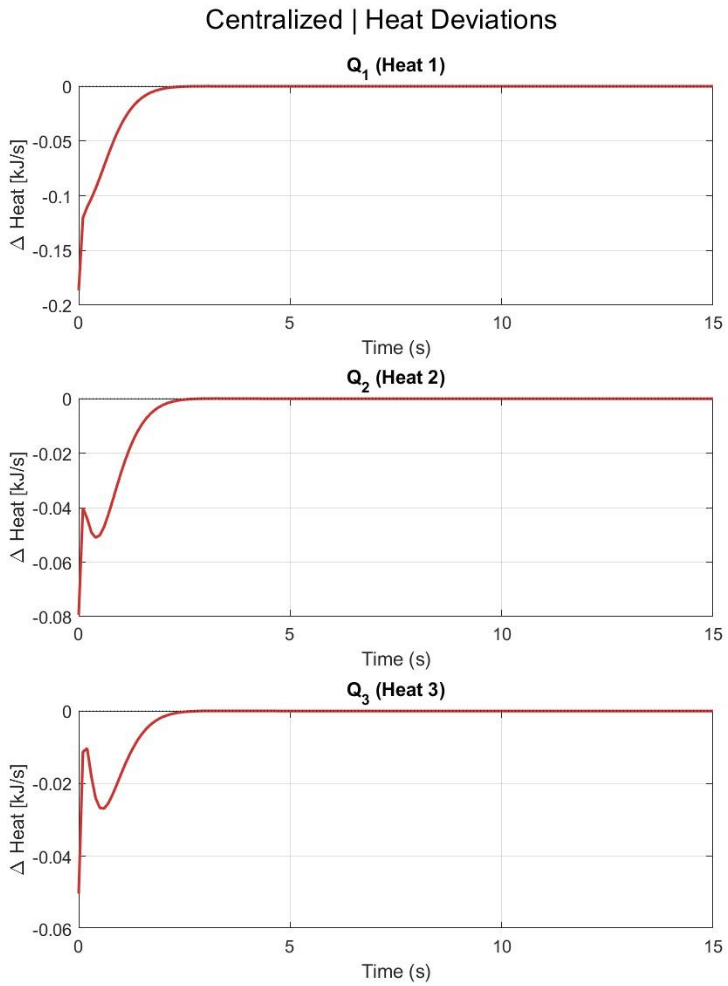
After adding damping, imaginary axis explosion is constrained



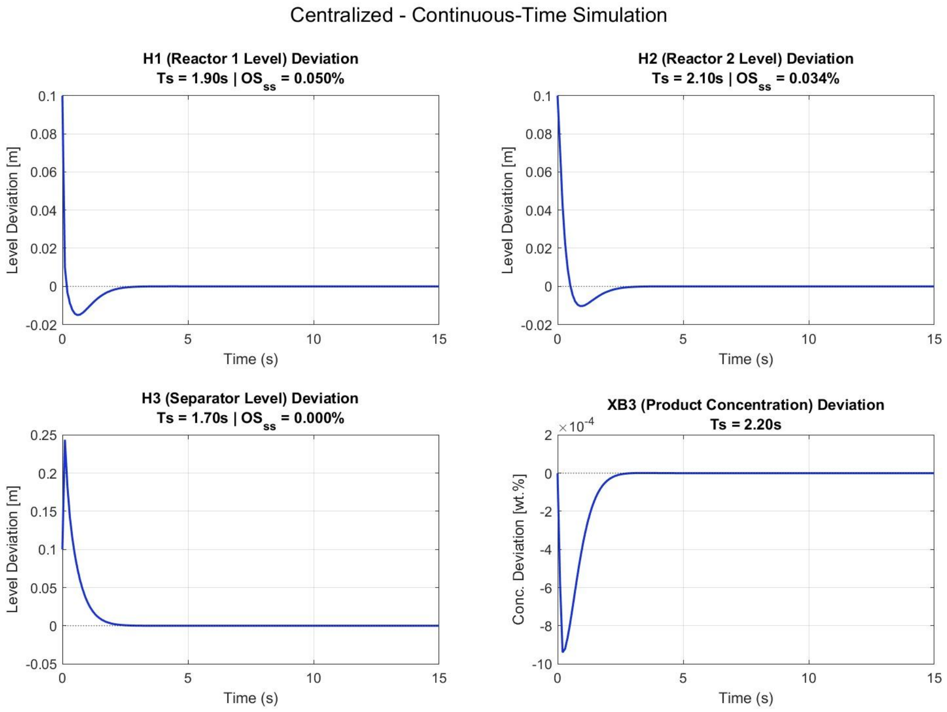


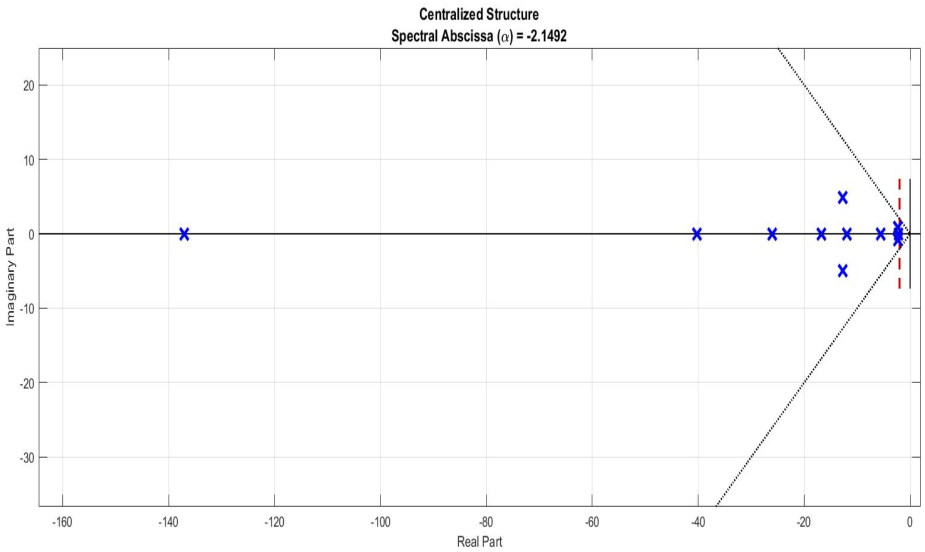
Higher control → lower deviation 📉

next: control minimization ⚙️



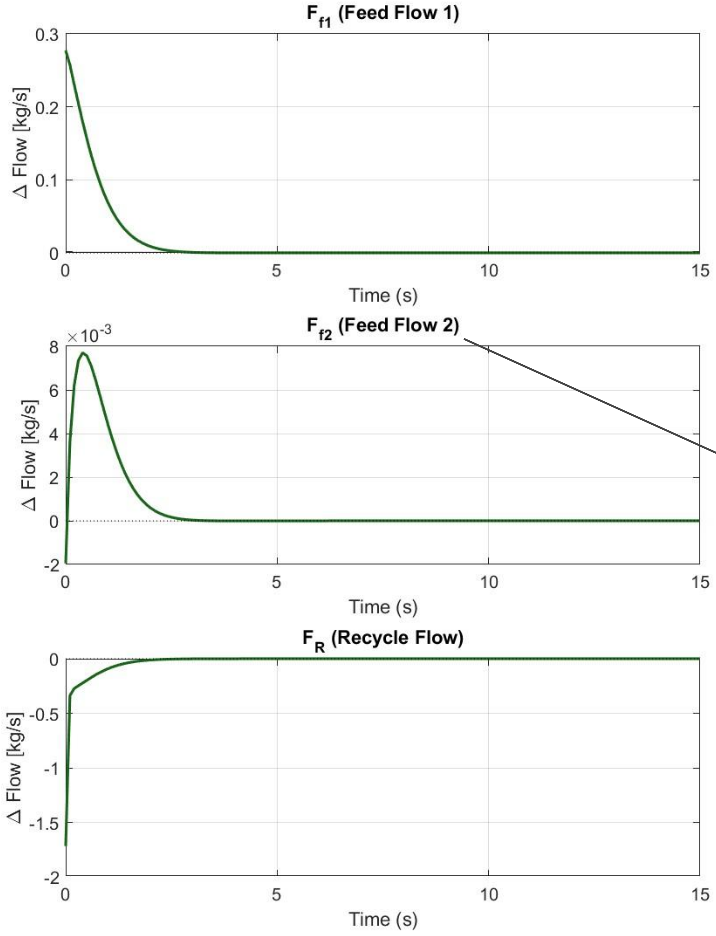
Continuous-Time , Control Effort Minimization





$$\|K_x\| = 1469.8846$$

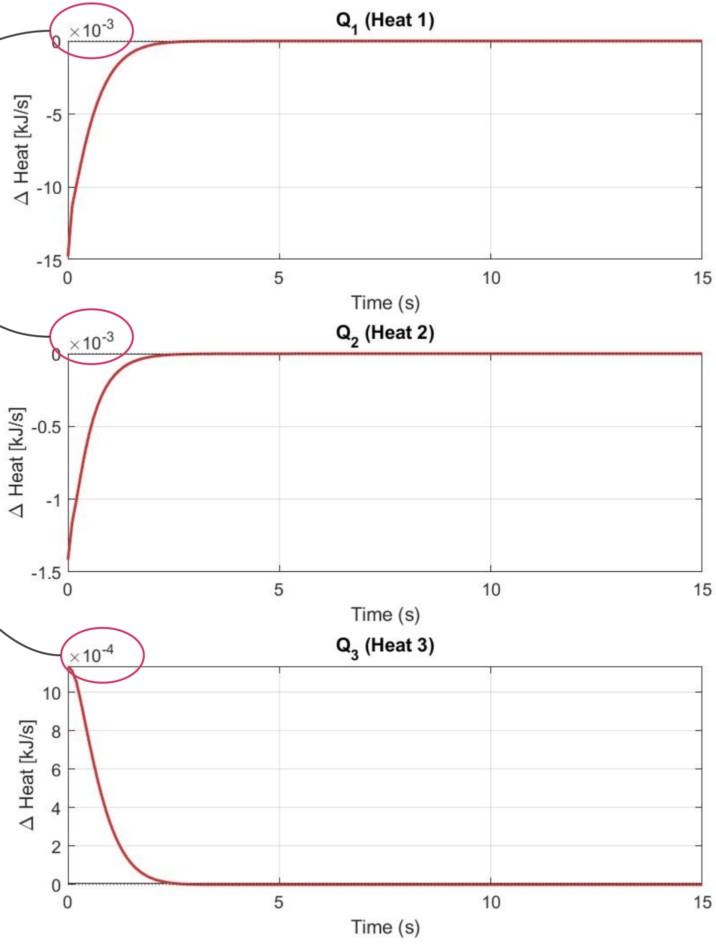
Centralized | Flow Deviations



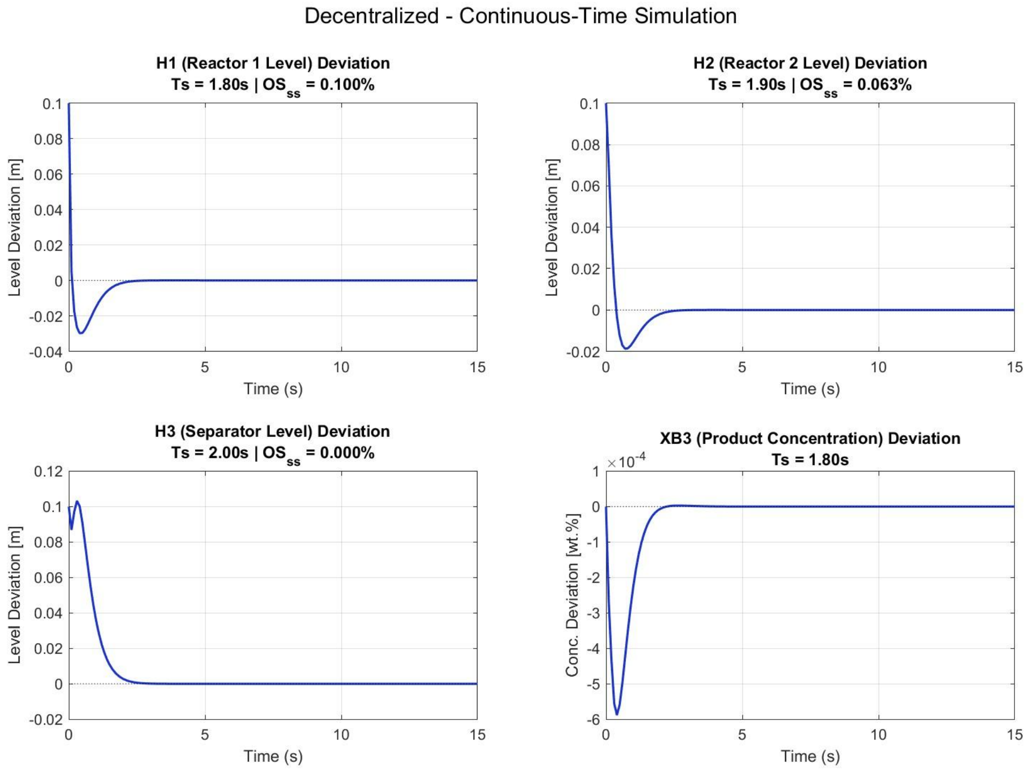
very small

❗ more effect here

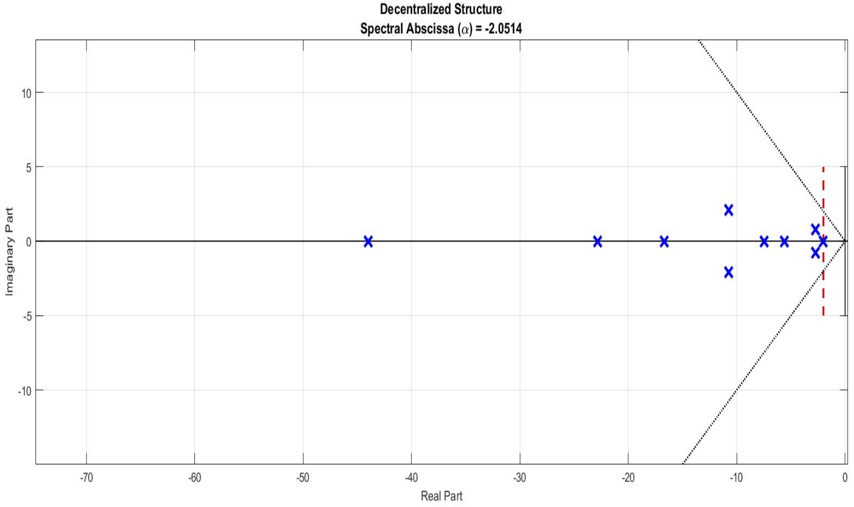
Centralized | Heat Deviations



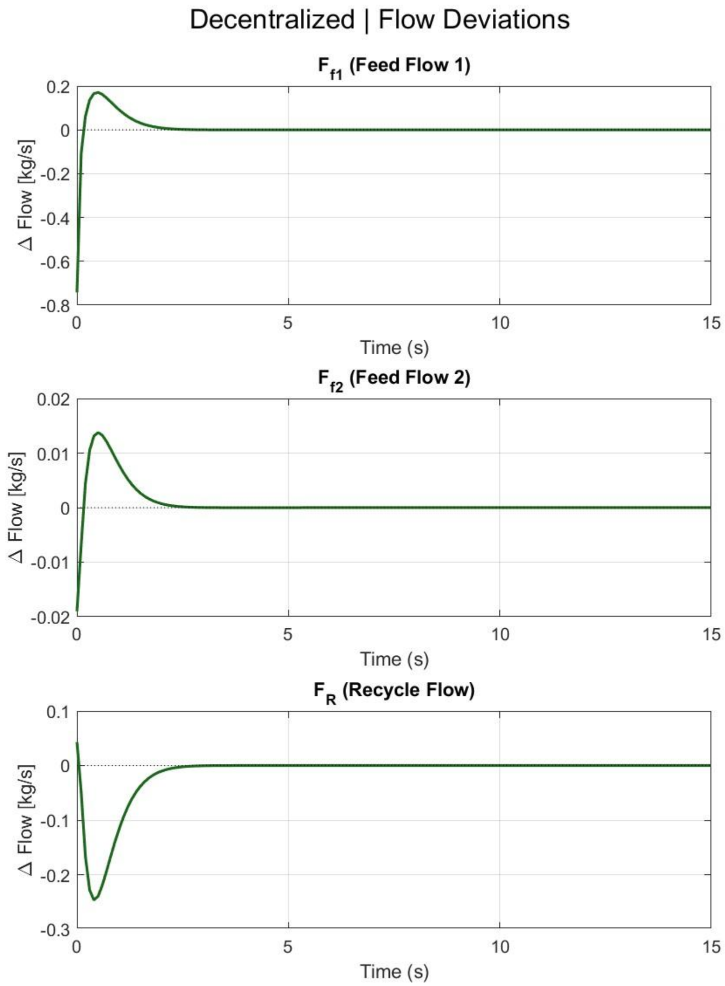
DECENTRALIZED



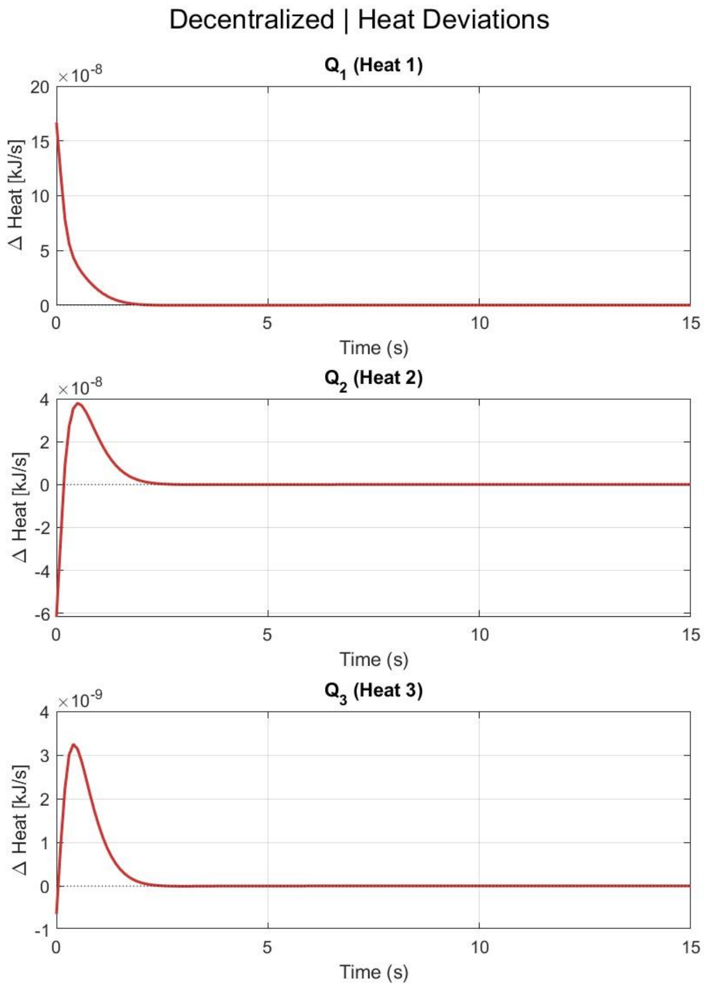
- H1, H2 Ts improved
- Slower H3



$$||K_x|| = 356.9604$$

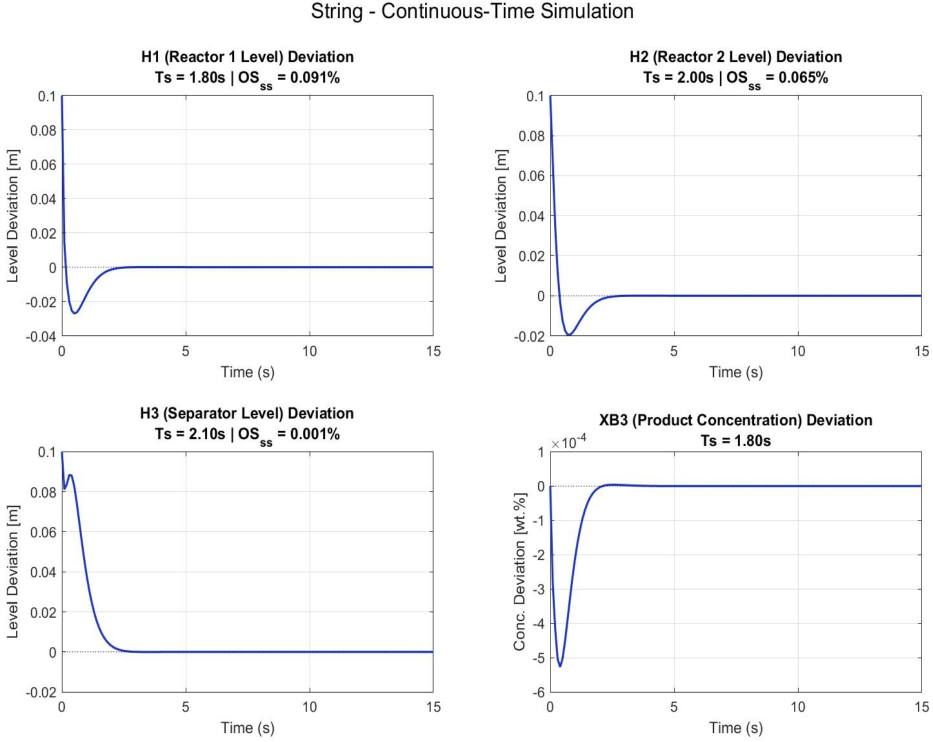


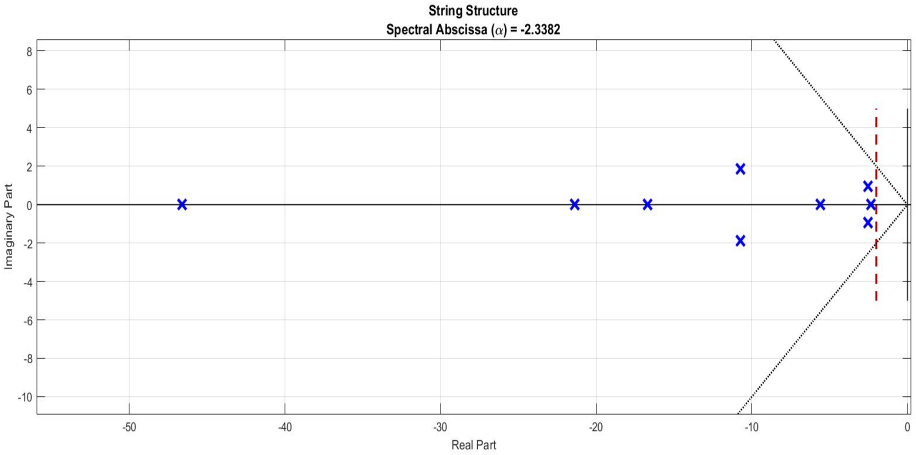
Decentralized: more effect for heat deviation



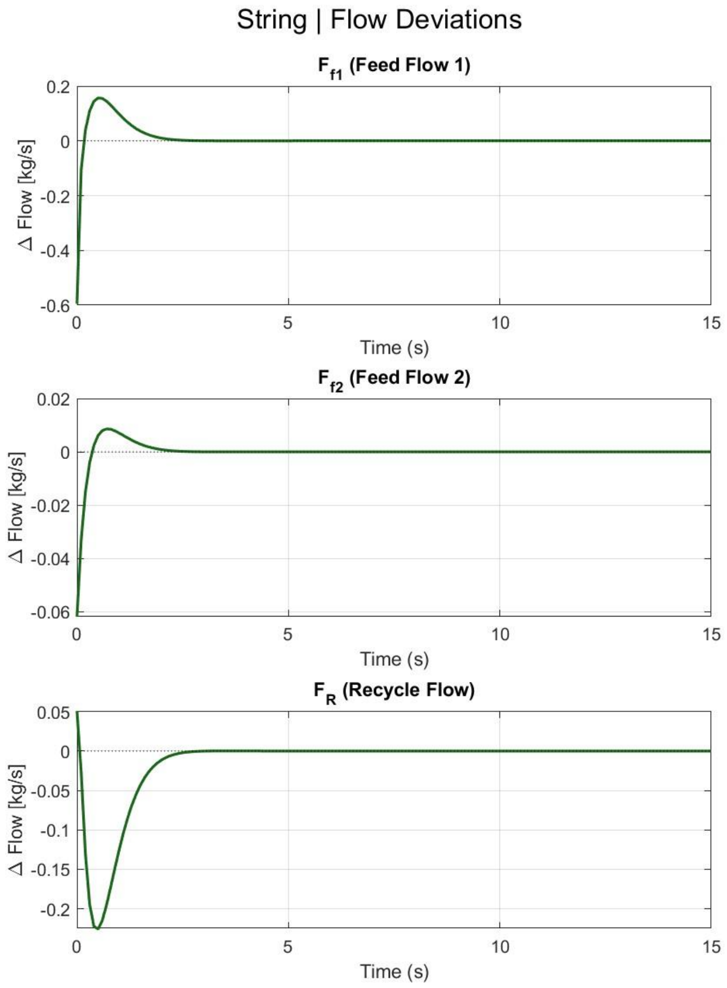
STRING

String Topology: Middle in performance

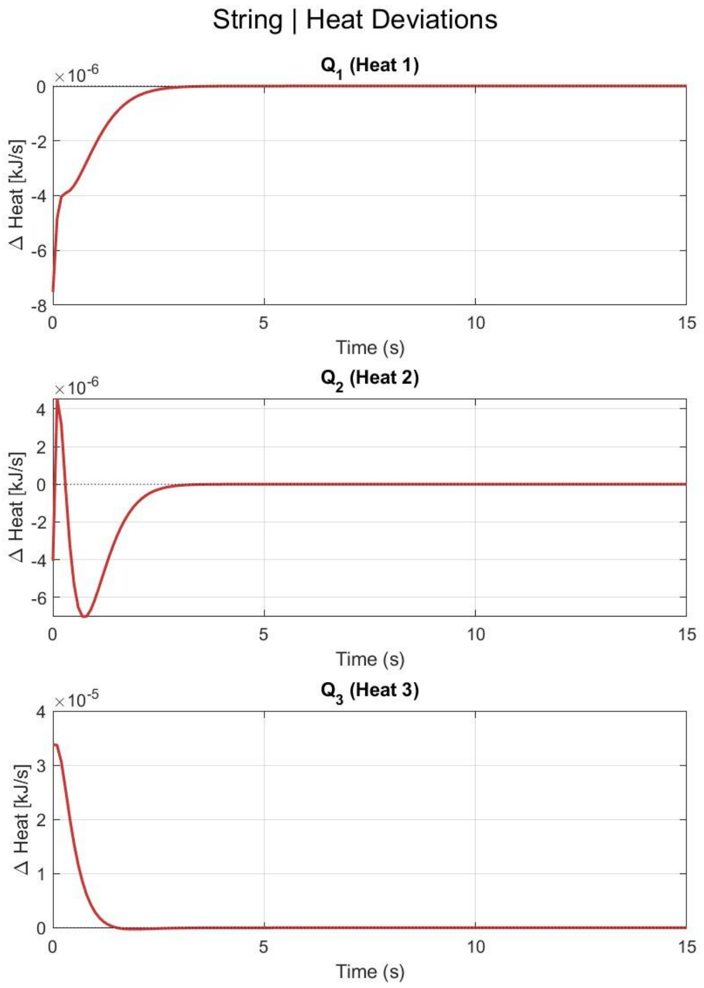




$$\|K_x\| = 530.3080$$



Flow: slightly more change
More deviation than centralized case



H2-Optimal Control

✅ **Using H-2 Control-> Suppresses undershoot** in the recycle flow Fr
✅ **Improves robustness** against external disturbances
✅ **Ensures stability** under strong dynamic coupling
✅ **Reduces control effort**, limiting actuator wear
✅ **Lowers energy consumption** while maintaining performance

$$J = trace(S)$$
$$s.t.$$

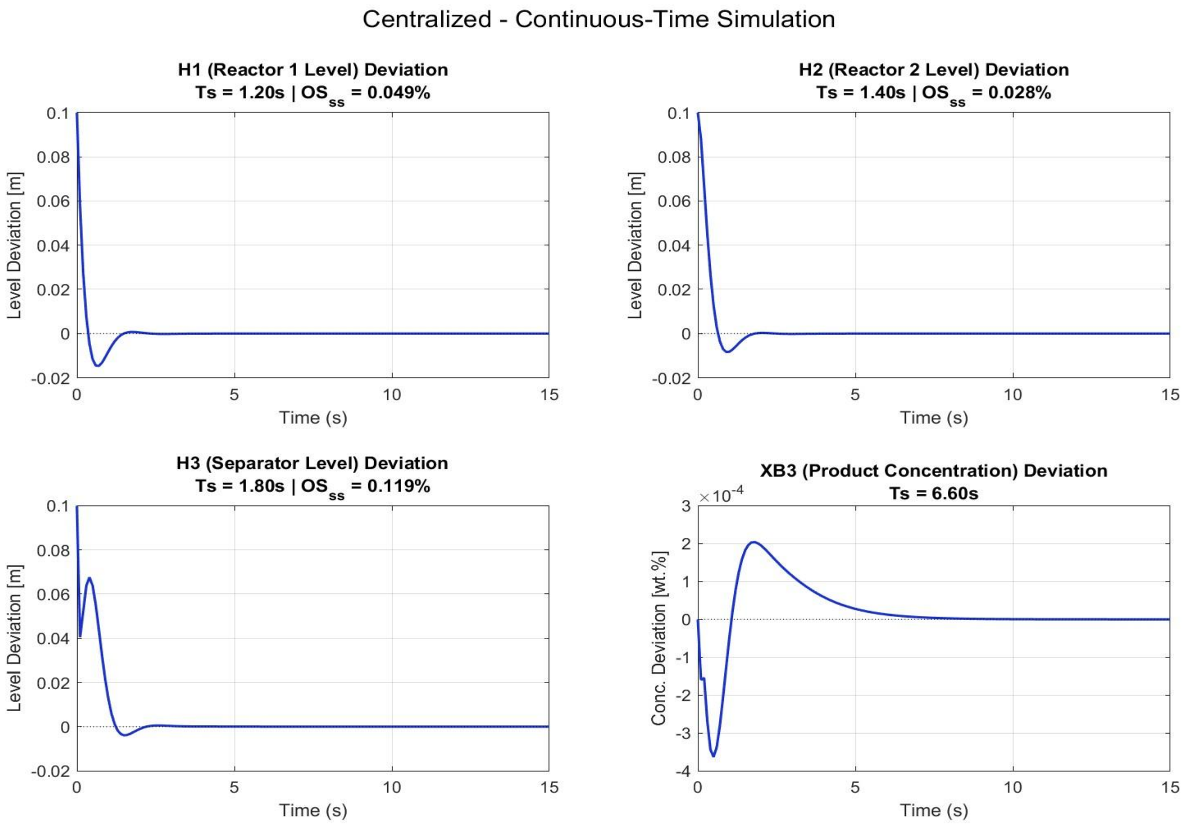
$$YA^T + AY + BL + L^T B^T + B_w B_w^T < 0\,, Y > 0$$

$$\begin{bmatrix} S & CY + D_u L \\ (CY + D_u L)^T & Y \end{bmatrix} > 0$$

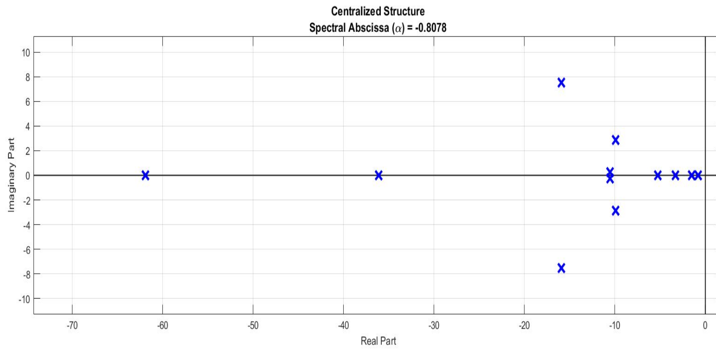
$$Q_{y_{1,2}} = diag(1,0,0,0.1)$$
$$Q_{y_3} = diag(1,0,1000,0)$$
$$R = 0.01\, eye(6)$$

🔺 ($F_D = 0.01 F_R$)

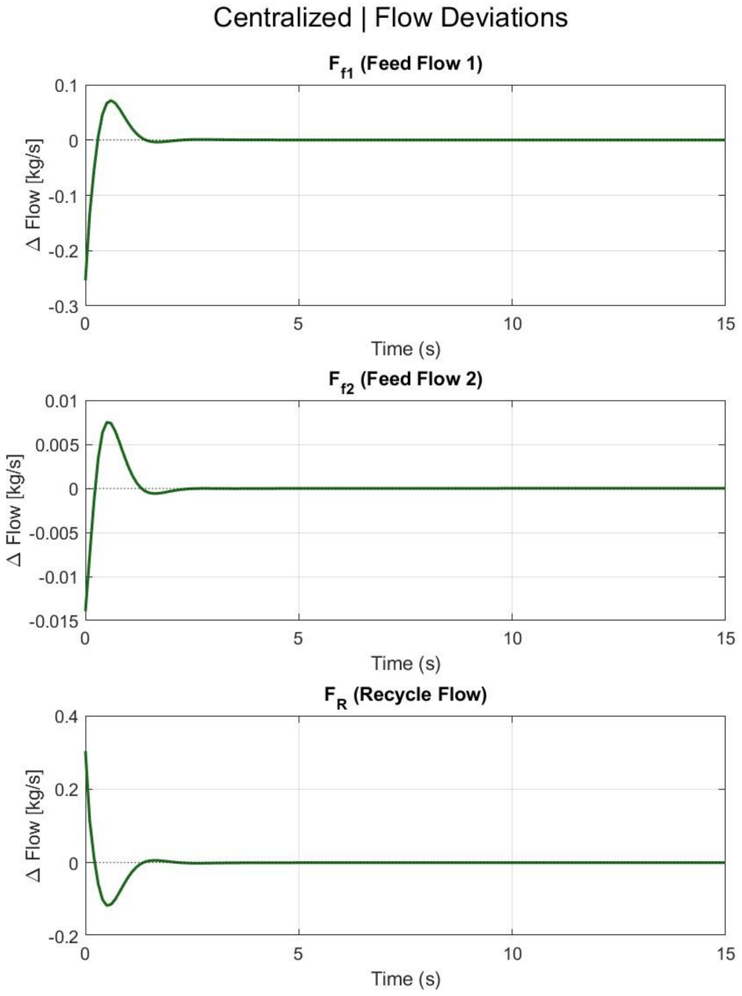
CENTRALIZED



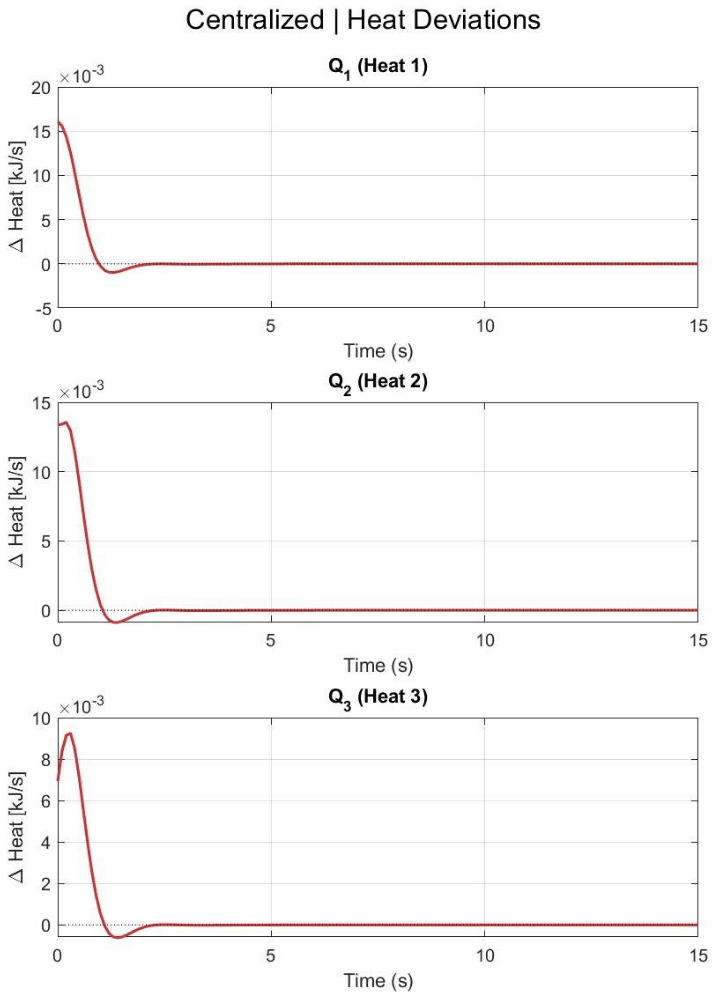
📈 Improved Ts



$$\|K_x\| = 12.9609$$

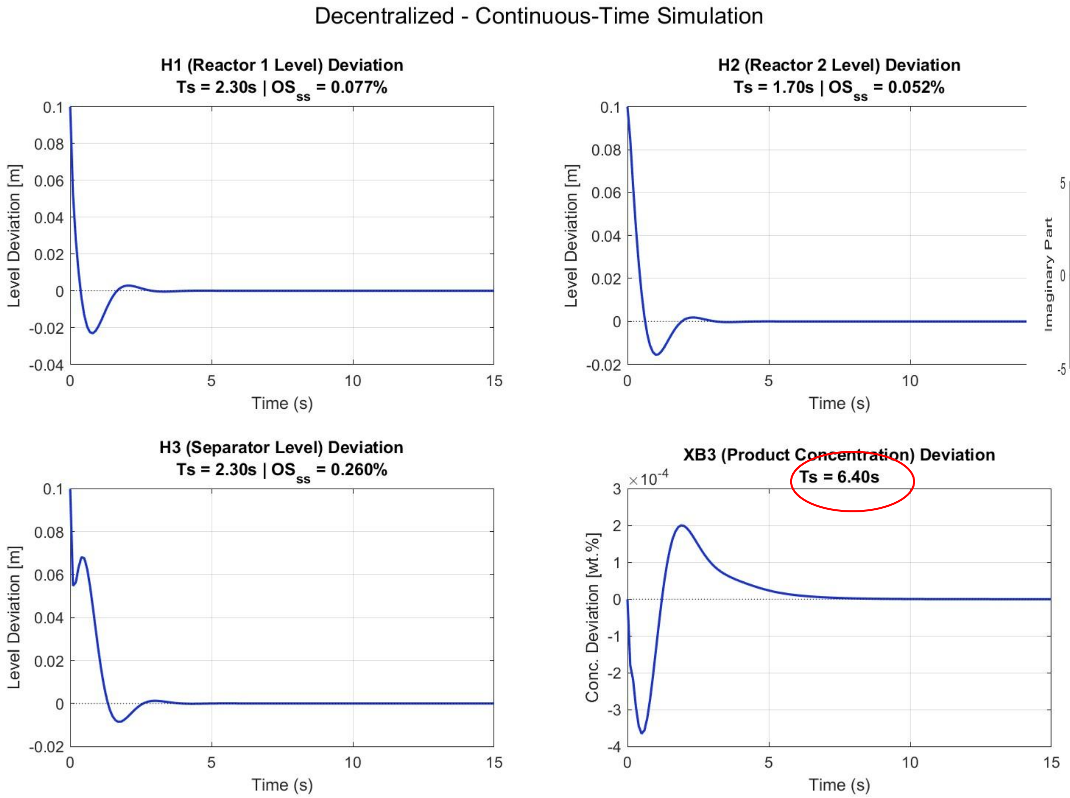


Centralized | Flow Deviations
F_{f1} (Feed Flow 1)
F_{f2} (Feed Flow 2)
F_R (Recycle Flow)
Δ Flow [kg/s]
Time (s)

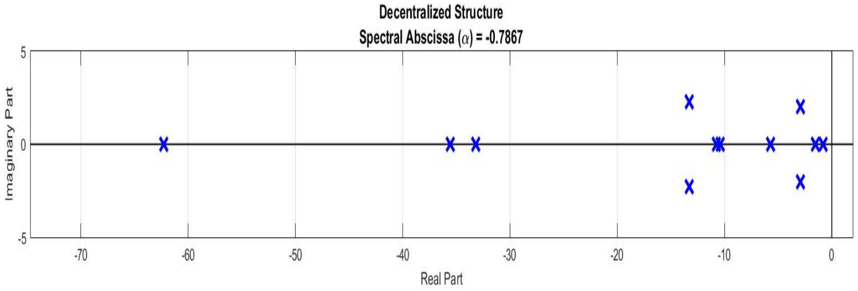


Centralized | Heat Deviations
Q_1 (Heat 1)
Q_2 (Heat 2)
Q_3 (Heat 3)
$\times 10^{-3}$
Δ Heat [kJ/s]
Time (s)

DECENTRALIZED

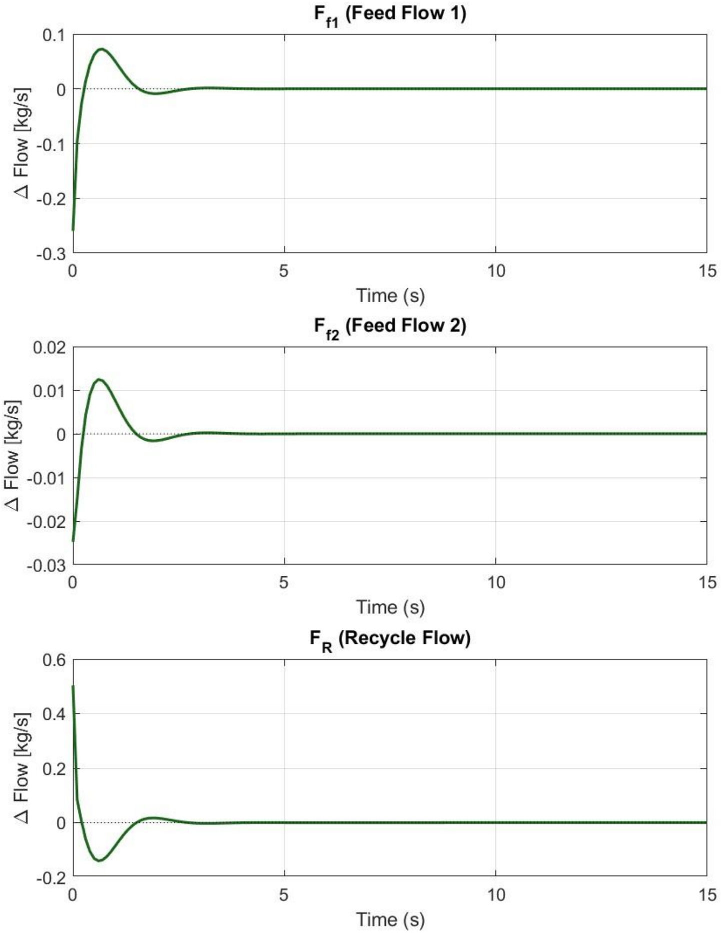


$$\|K_x\| = 1.8777$$

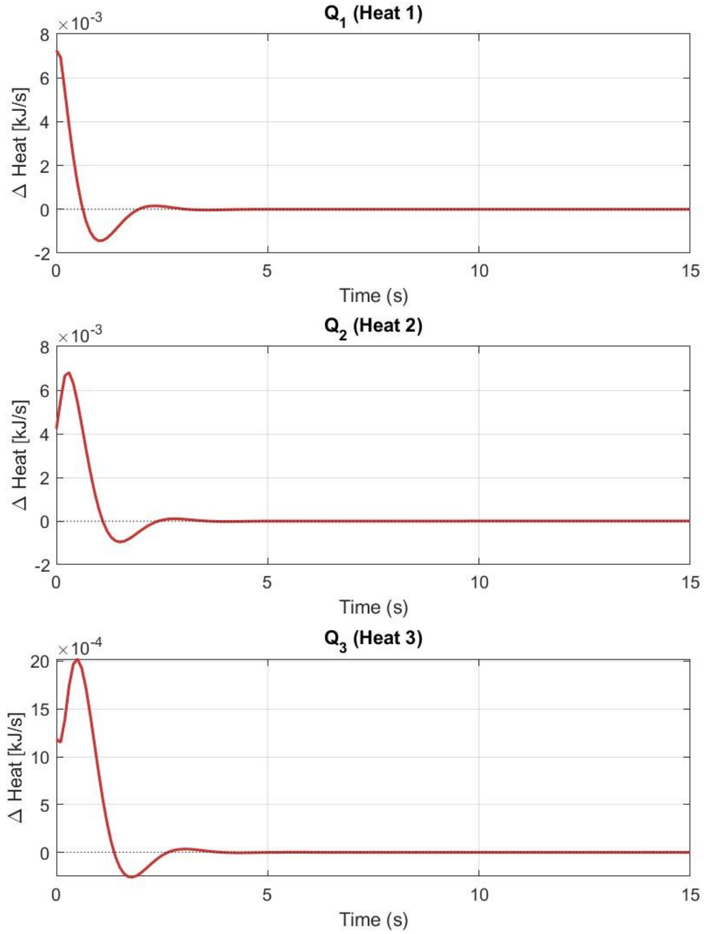


H-2 did not improve settling time for XB3!

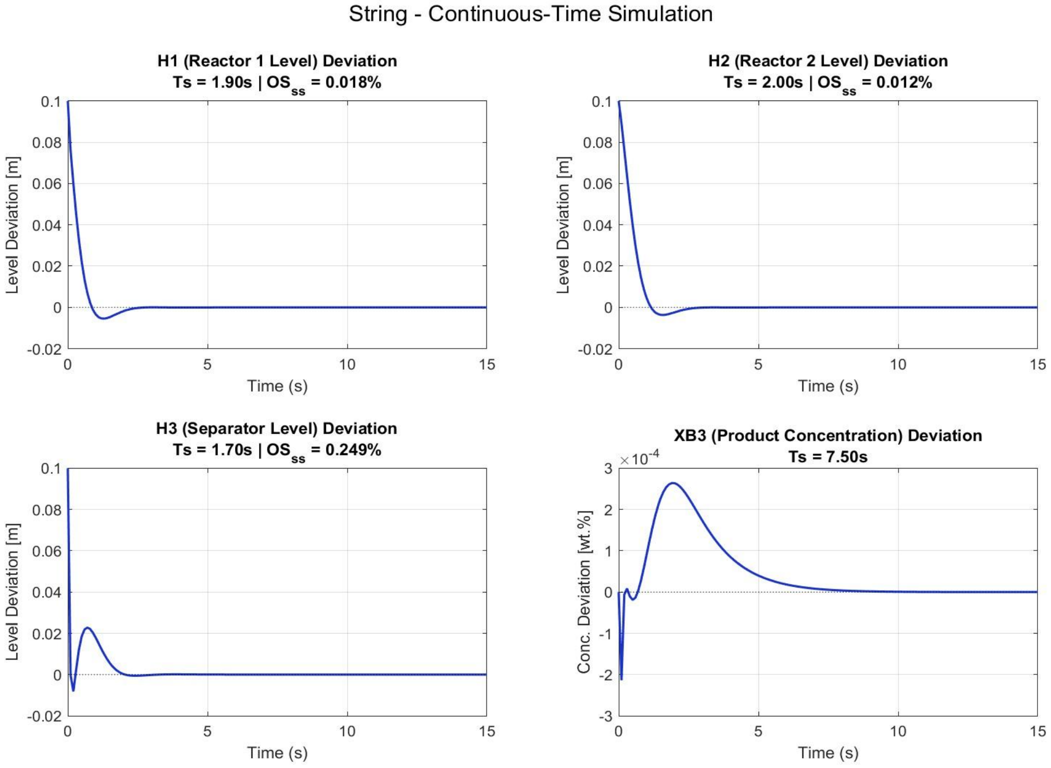
Decentralized | Flow Deviations

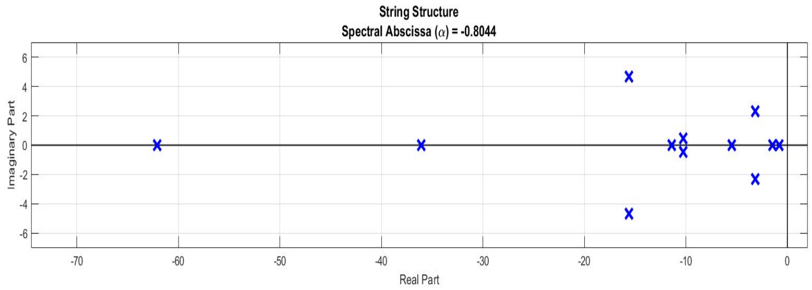


Decentralized | Heat Deviations

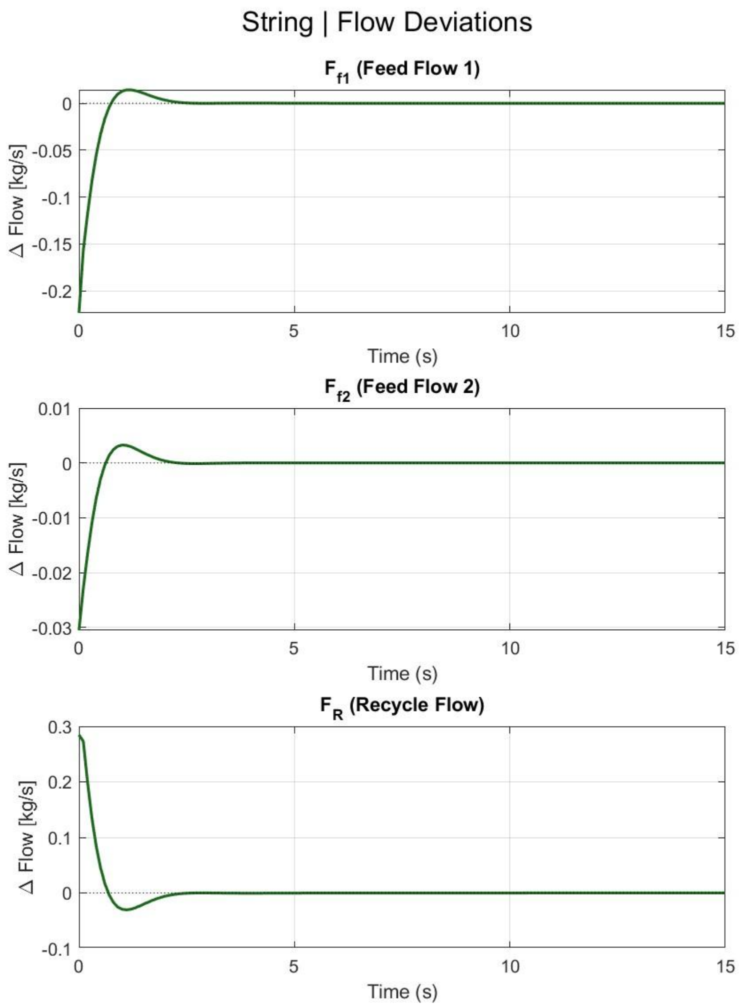


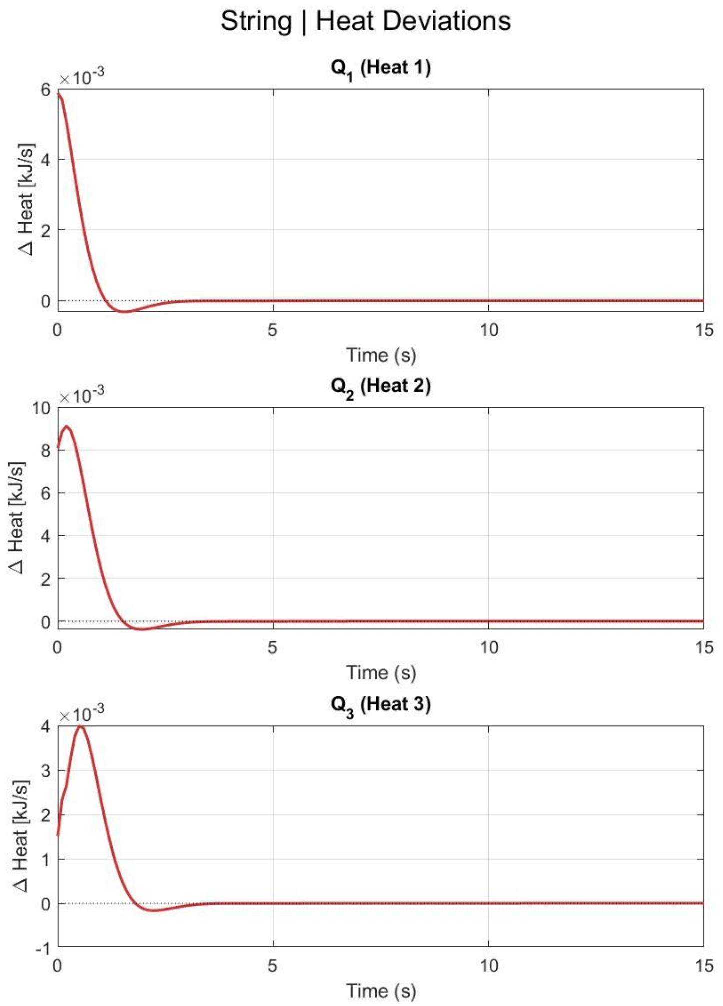
STRING





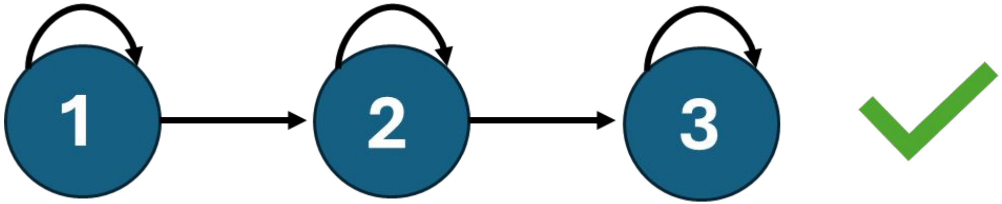
$$\|K_x\| = 3.3456$$


String | Flow Deviations
F f1 (Feed Flow 1)
F f2 (Feed Flow 2)
F R (Recycle Flow)
Δ Flow [kg/s]
Time (s)


String | Heat Deviations
Q 1 (Heat 1)
Q 2 (Heat 2)
Q 3 (Heat 3)
Δ Heat [kJ/s]
×10-3
Time (s)

CONCLUSIONS

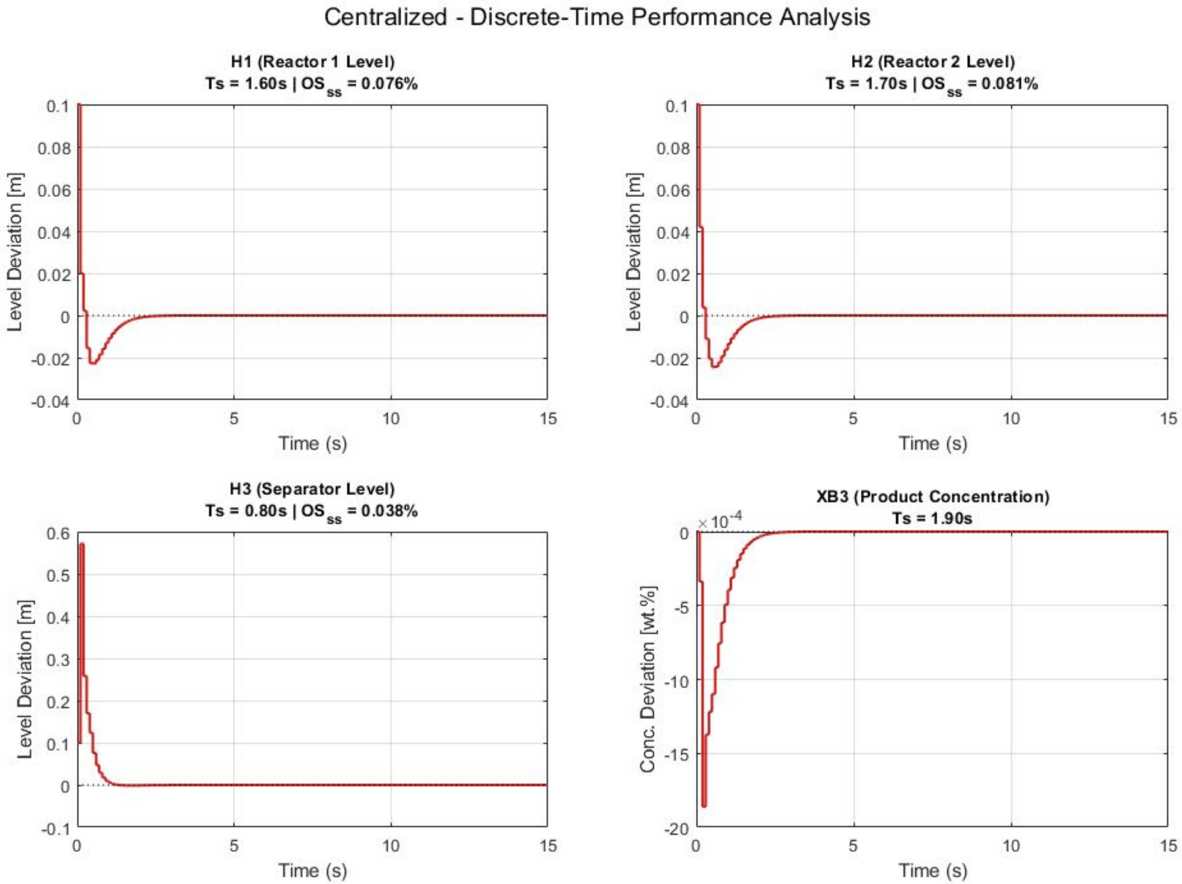
- **Dynamic Response and Trade-offs**: While the settling times for levels H_1, H_2 and H_3 slowed by approximately 0.7–1 second compared to the open-loop response, this sacrifice in speed enabled a significant 5-second gain in the control of the critical product quality variable, X_{B3}
- **Constraint Satisfaction:** All strict input constraints were successfully maintained without violation, with specific emphasis on the feed flow F_{f2} ensuring the process remained within safe physical and operational limits at all times.
- **Distributed Topology Preference:** Due to the coupled dynamics inherent in the reactor-separator sequence, a String Distributed control structure is preferred as it preserves the flexibility of independent unit operations while managing plantwide interactions.



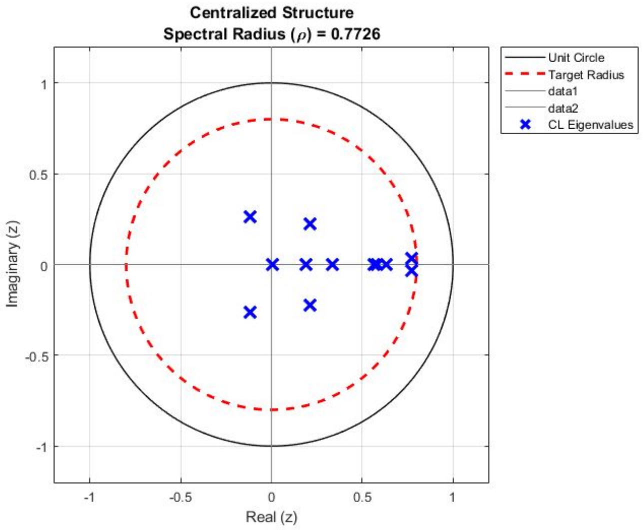
DISCRETE TIME CONTROL

Settling Time	Overshoot	Control Effort	Input Constraints
$T_s \cong -\dfrac{5}{ln(\rho)}$ $e^{Re(\lambda)T_s} \cong 0.8 = \rho$	$\lambda(F_{cl}) \in \{z : \; \|z - c\| \leq r\}$ $z = c + re^{j\theta} \;\Rightarrow\; \|\Im(z)\| \leq r$	$J = \alpha_L K_L + \alpha_Y K_Y$ $\alpha_L = 0.1 \,, \alpha_Y = 10$	$\mu = \min(u_{\max} - u_{ss} \,, u_{ss} - u_{min})$ $(K_i x_i)^2 \leq \mu_i^2, where\ L = KY$ $\mu_i^2 - L_i Y^{-1} L_i^T \geq 0$
$\begin{bmatrix} \rho^2 P - FPF^T - FL^T G^T - GLF^T & GL \\ L^T G^T & P \end{bmatrix} > 0$	$\begin{bmatrix} r_{damp}^2 P - \tilde{F}P\tilde{F}^T - \tilde{F}L^T G^T - GL\tilde{F}^T & GL \\ L^T G^T & P \end{bmatrix} > 0$ $\tilde{F} = F - \alpha I$	• $\begin{bmatrix} K_L I & L^T \\ L & I \end{bmatrix} \geq 0$ • $\begin{bmatrix} K_Y I & I \\ I & Y \end{bmatrix} \geq 0$	• $\begin{bmatrix} \mu^2 & L \\ L^T & Y \end{bmatrix} \geq 0$

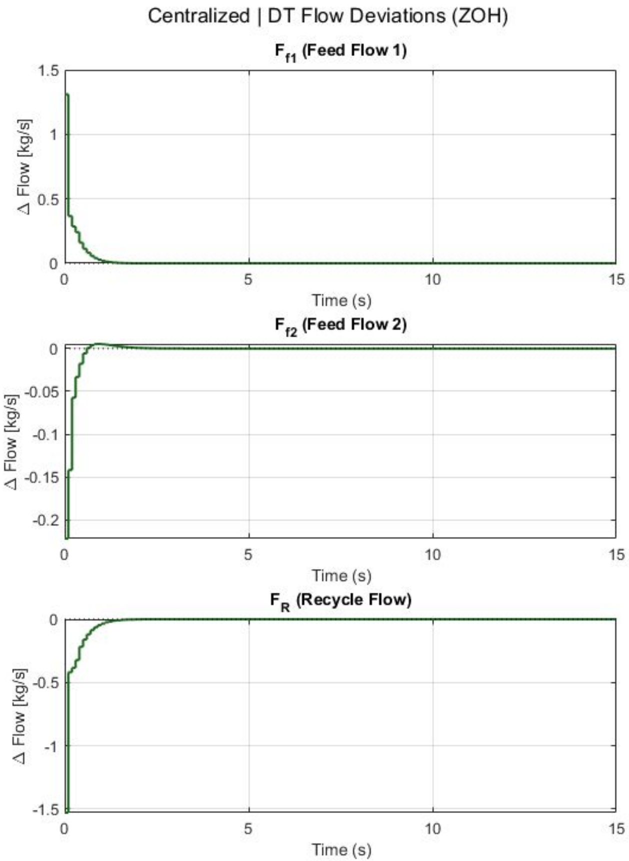
DT - CENTR - EFFORT MIN - RHO = 0.8 — (pg.1)

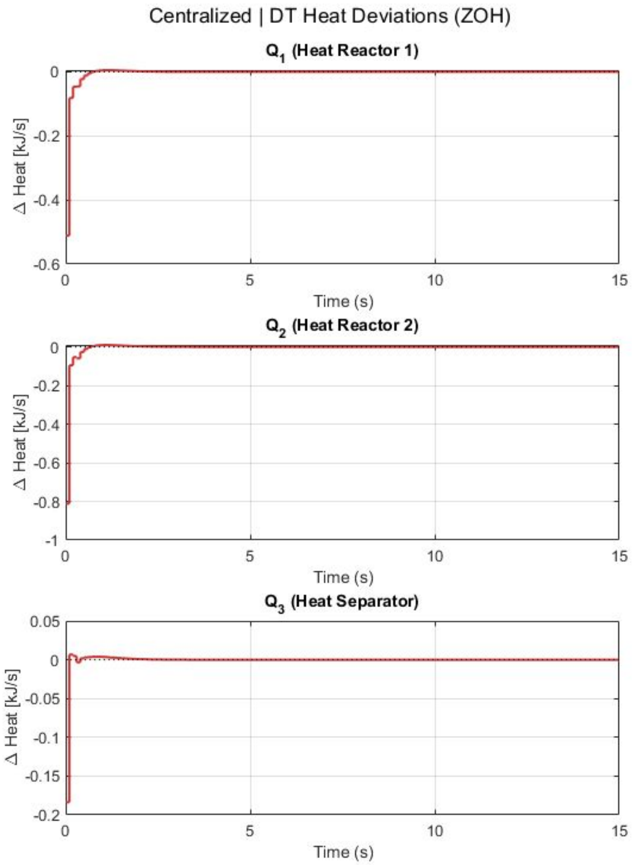


- Initial condition 0.1 for all levels.
- Level oscillations not bad.
- Poles on the left, and angled.



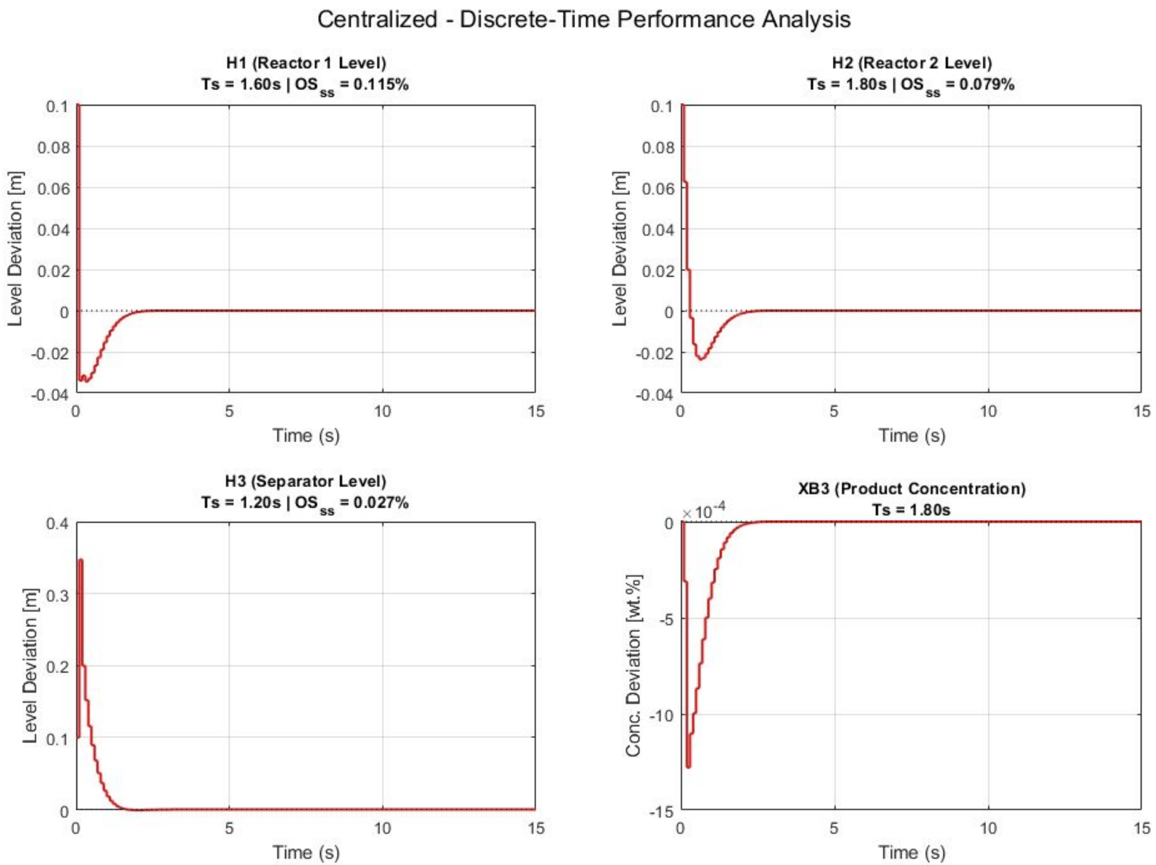
DT - CENTR - EFFORT MIN - RHO = 0.8 — (pg.2)



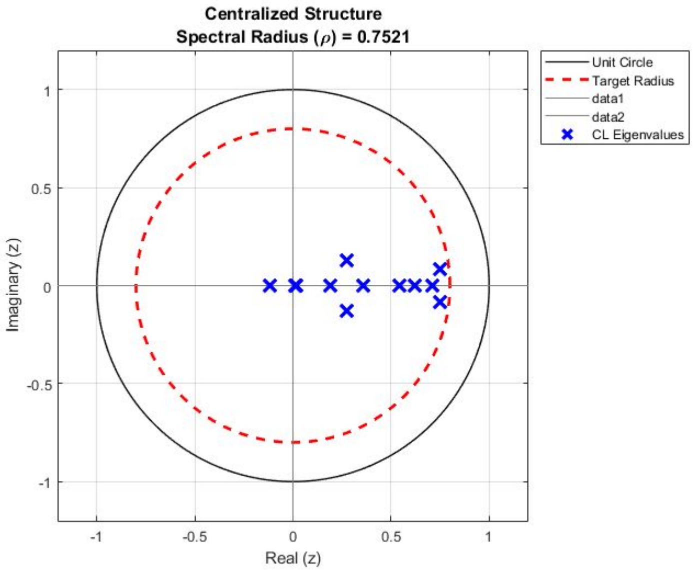


Even for this relaxed min. decay. condition, Ff1 goes over bound.

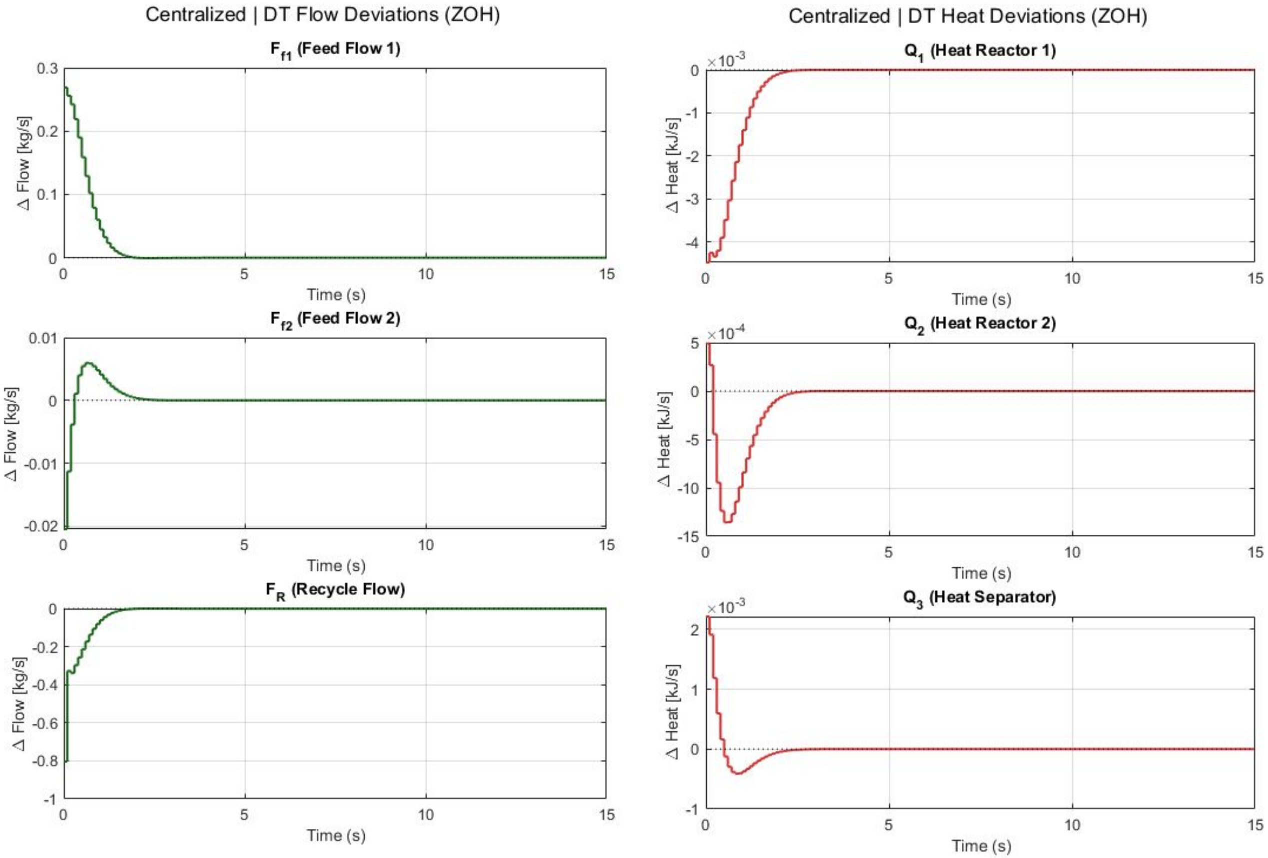
DT - CENTR - INPUT ELLIPSE - RHO = 0.8 — (pg.1)



- Very similar results.
- ||K|| = 1356
- Pole angles better, but similar oscillations on levels

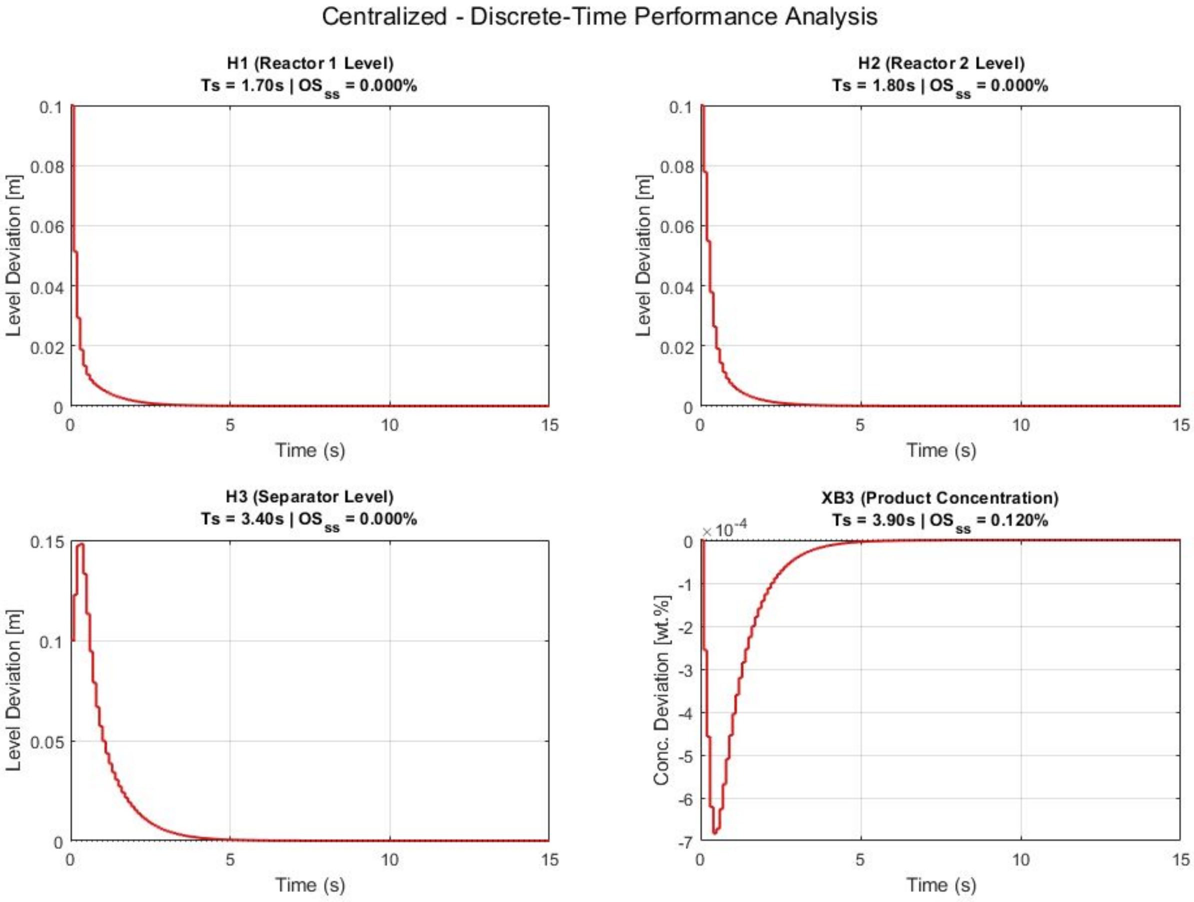


DT - CENTR - INPUT ELLIPSE - RHO = 0.8 — (pg.2)

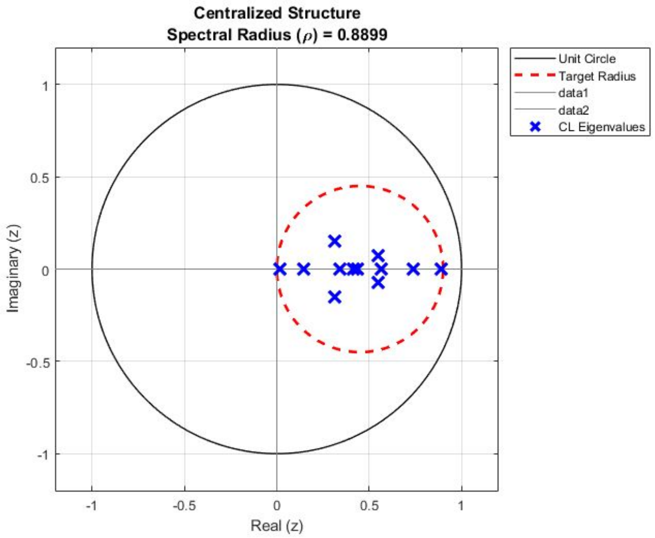


- Now input deviations within boundaries as desired.

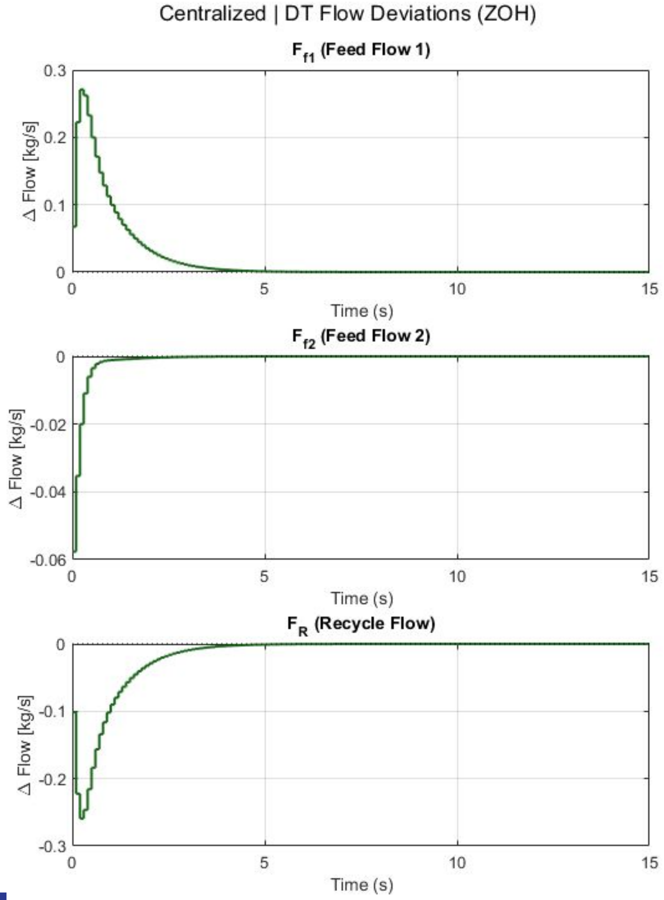
DT - CENTR - INPUT ELLIPSE - C = 0.45 - RHO = 0.45 — (pg.1)

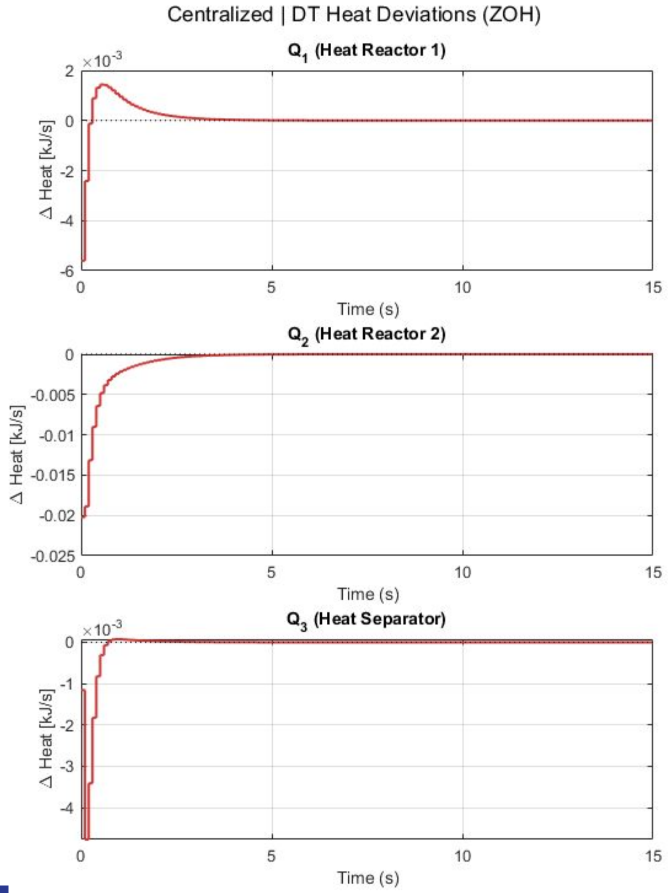


- $||K|| = 1334$
- Pole angles similar, but LHP poles avoided.
- OS improved, but min. decay rate lower. Shows on settling times.

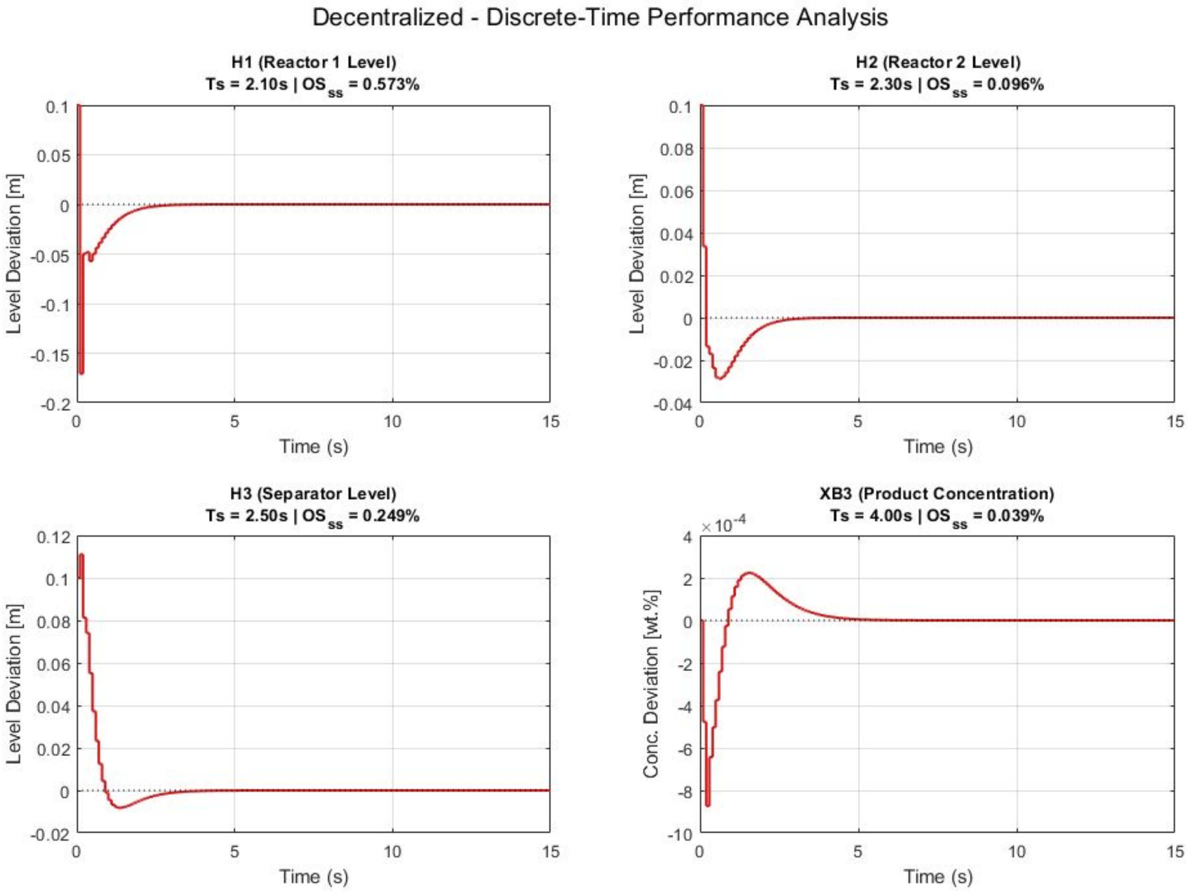


DT - CENTR - INPUT ELLIPSE - C = 0.45 - RHO = 0.45 — (pg.2)

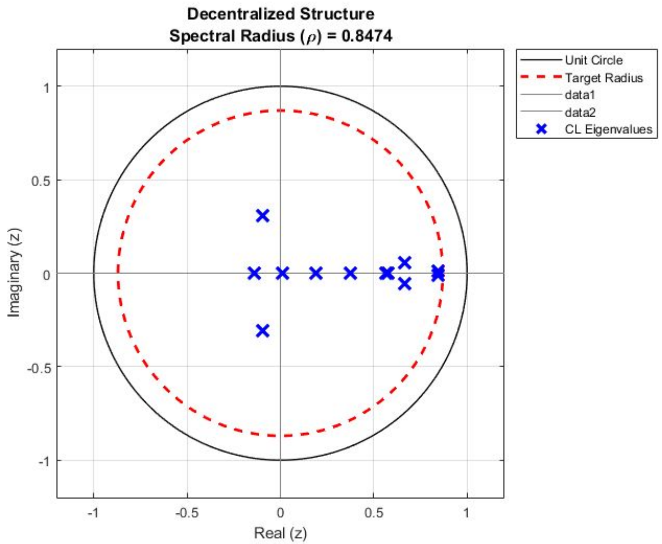




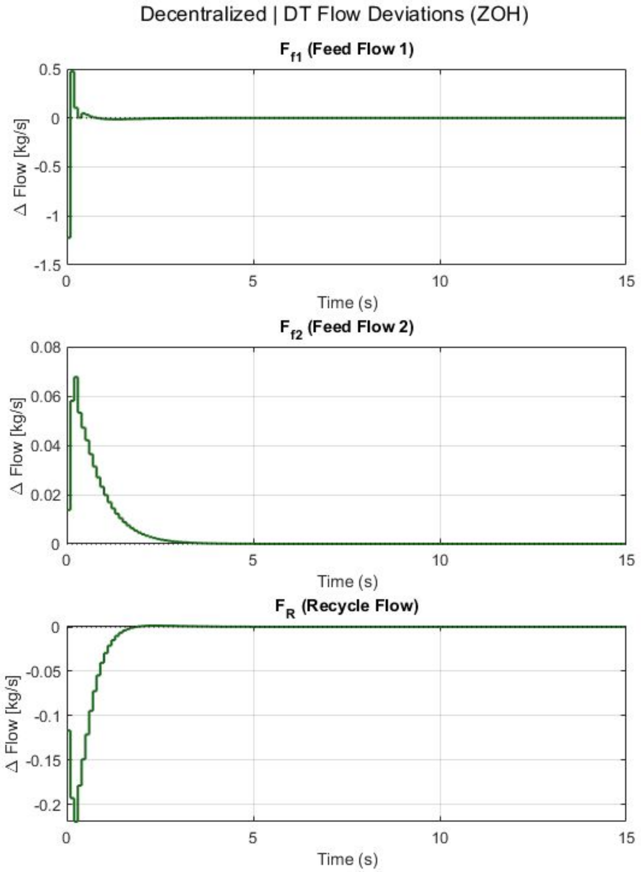
DT - DECENTR - INPUT ELLIPSE - RHO = 0.85 — (pg.1)

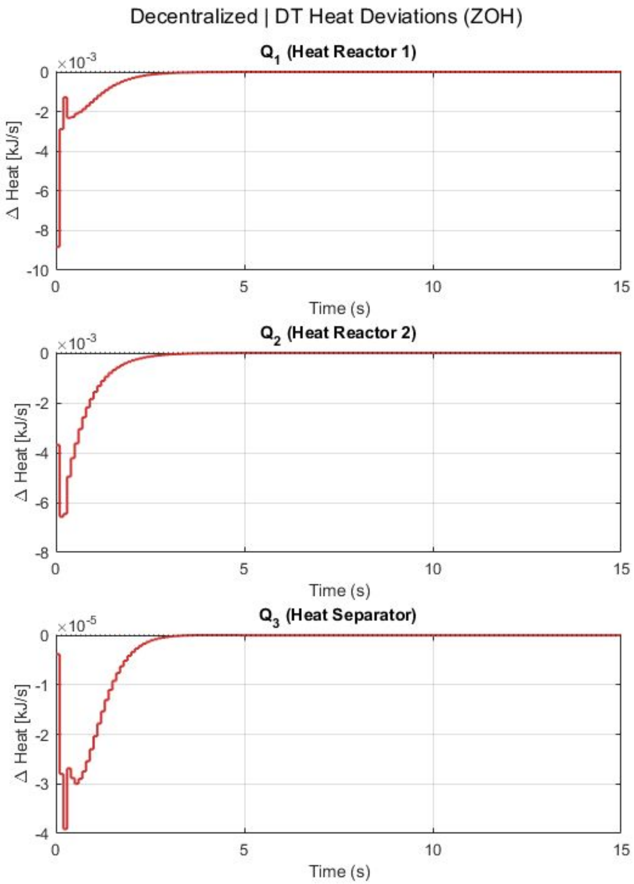


- Much harder to impose min. decay. rate in decentralised.
- ||K|| = 1486, more aggressive even for a more relaxed condition.



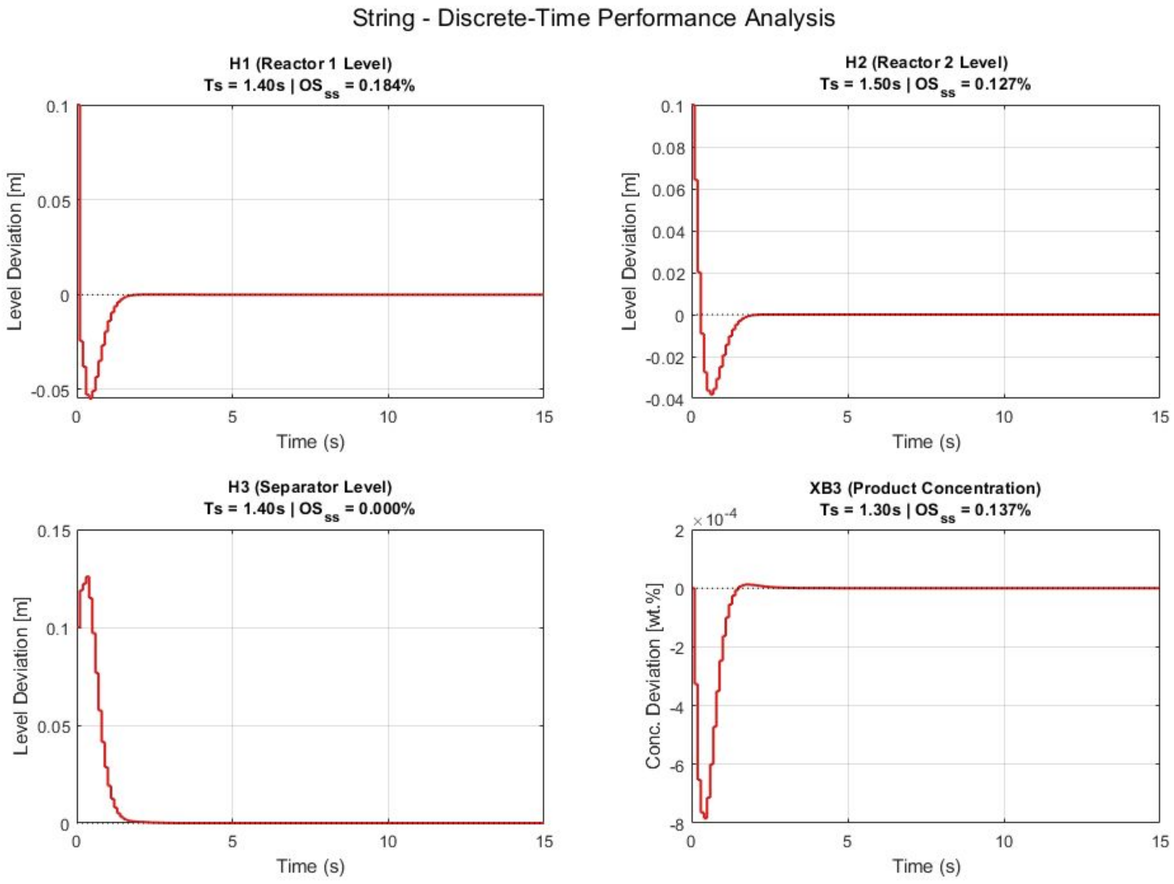
DT - DECENTR - INPUT ELLIPSE - RHO = 0.85 — (pg.2)



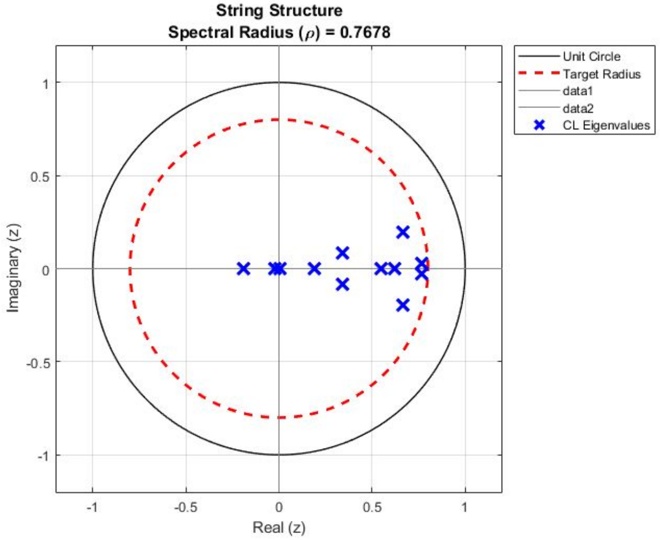


- More aggressive inputs observed.

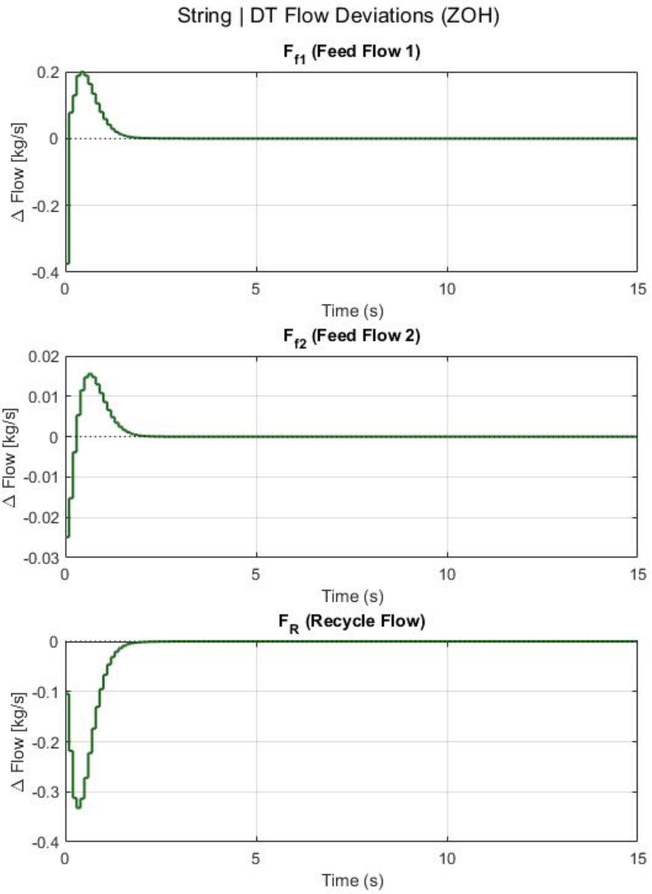
DT - STRING - INPUT ELLIPSE - RHO = 0.80 — (pg.1)

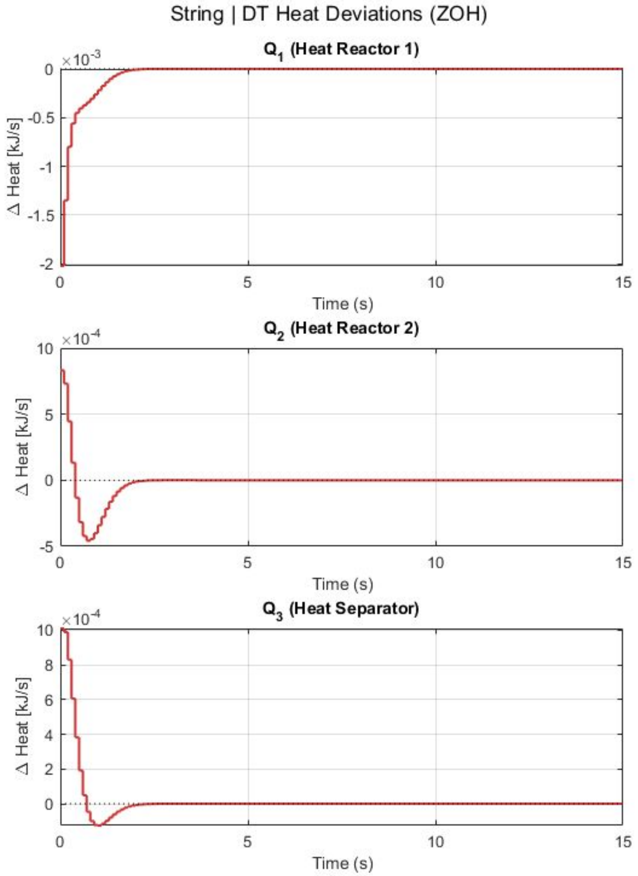


- Can achieve the centralized case's min. decay rate.
- ||K|| = 580, less aggressive than decentralised.



DT - STRING - INPUT ELLIPSE - RHO = 0.80 — (pg.2)





- Similar to the centralized case again.
- Additionally, can't force tighter disk constraints in the string topology, unlike in the centralized case.

H2-NORM MINIMIZATION WITH LMIs

$$\begin{aligned} x_{k+1} &= Fx_k + Gu_k + G_w w_k \\ z_k &= Hx_k + D_u u_k \end{aligned}$$

Choosing LQ-like performance: $H = \begin{bmatrix} \sqrt{Q} \\ 0 \end{bmatrix}, D_u = \begin{bmatrix} 0 \\ \sqrt{R} \end{bmatrix}$

$$\min_{Y,S,L} trace(S)$$

$$L = K_x P$$

$$s.t.$$

$$\begin{bmatrix} P - FPF^T - FL^T G^T - GLF^T - G_w G_w^T & GL \\ L^T G^T & P \end{bmatrix} \geq 0$$

$$\begin{bmatrix} S & HP + D_u L \\ L^T D_u^T + PH^T & P \end{bmatrix} \geq 0$$

Partitioned weight matrices:

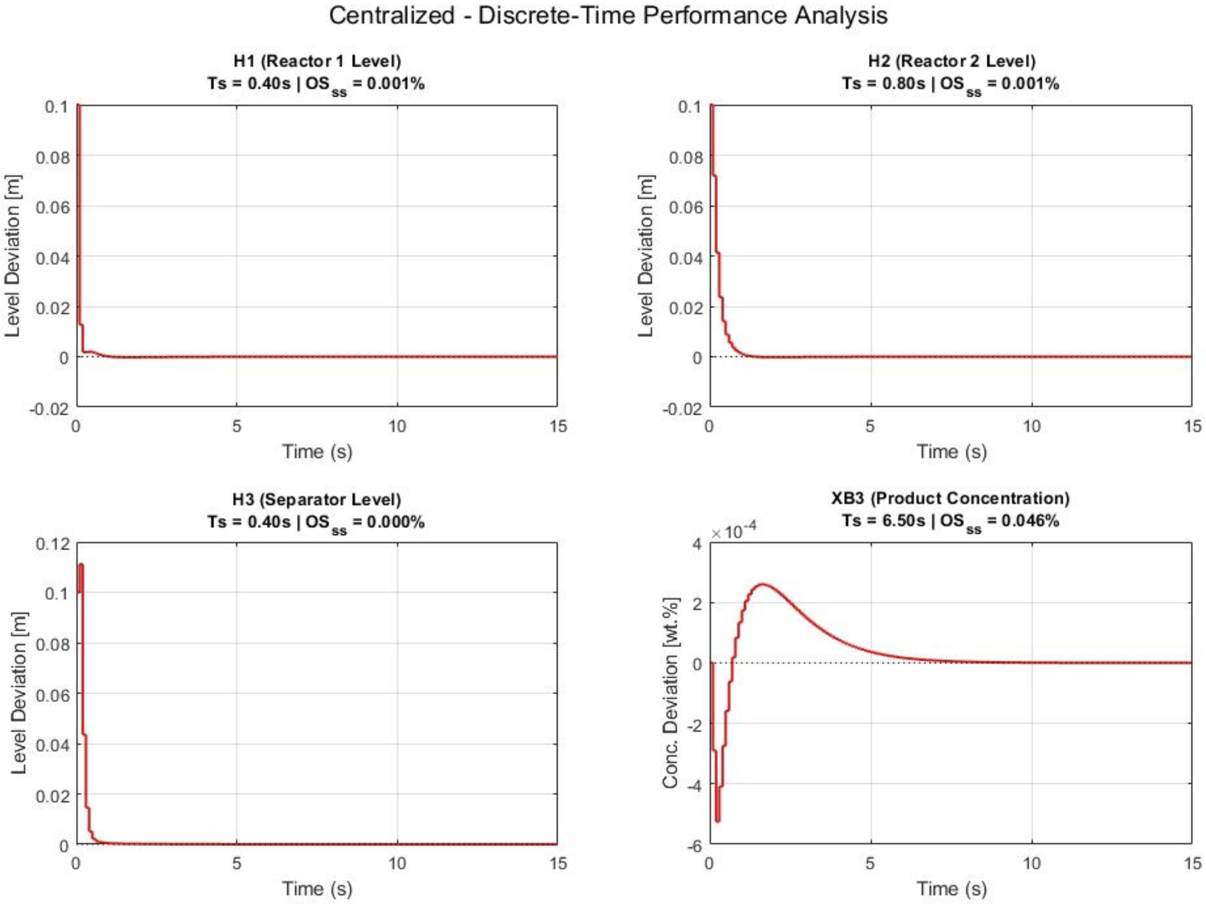
$$Q_{y_1} = diag(1, 0, 0, 0.1)$$

$$Q_{y_2} = diag(1, 0, 0, 0.1)$$

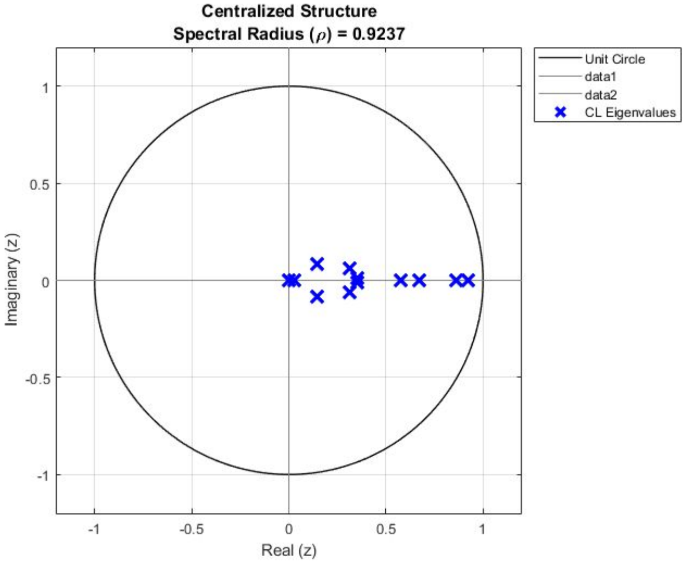
$$Q_{y_3} = diag(1, 0, 10^3, 0)$$

$$R_1 = R_2 = R_3 = 0.01I$$

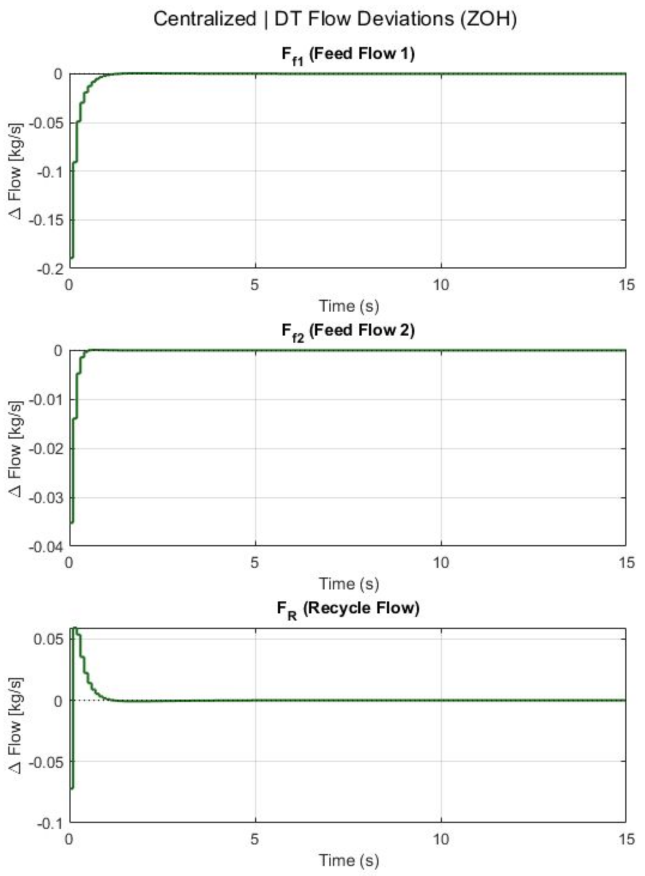
DT - CENTR - INPUT ELLIPSE - H2 MINIMIZATION – (pg.1)

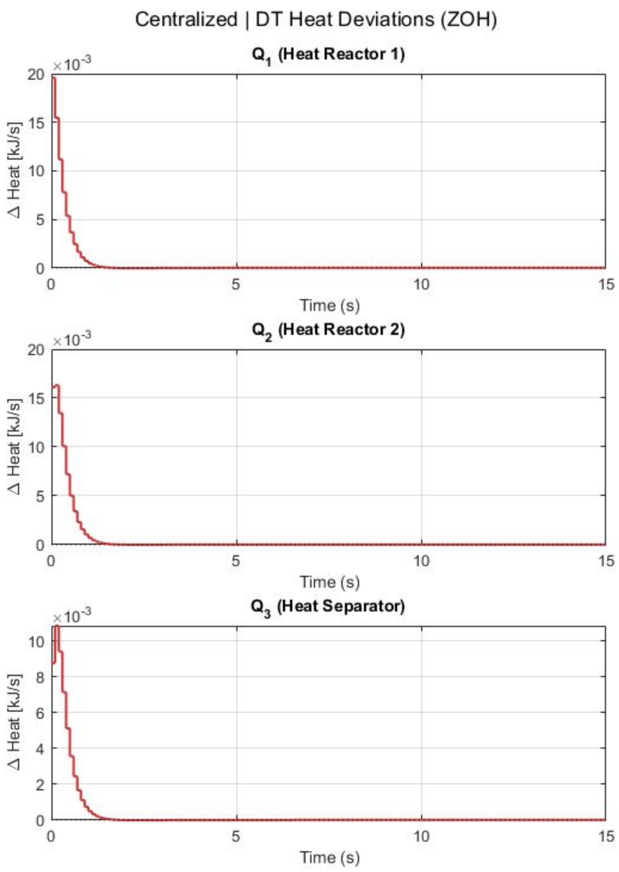


- Faster settling times on all inspected states compared to open loop.
- ||K|| = 5.26

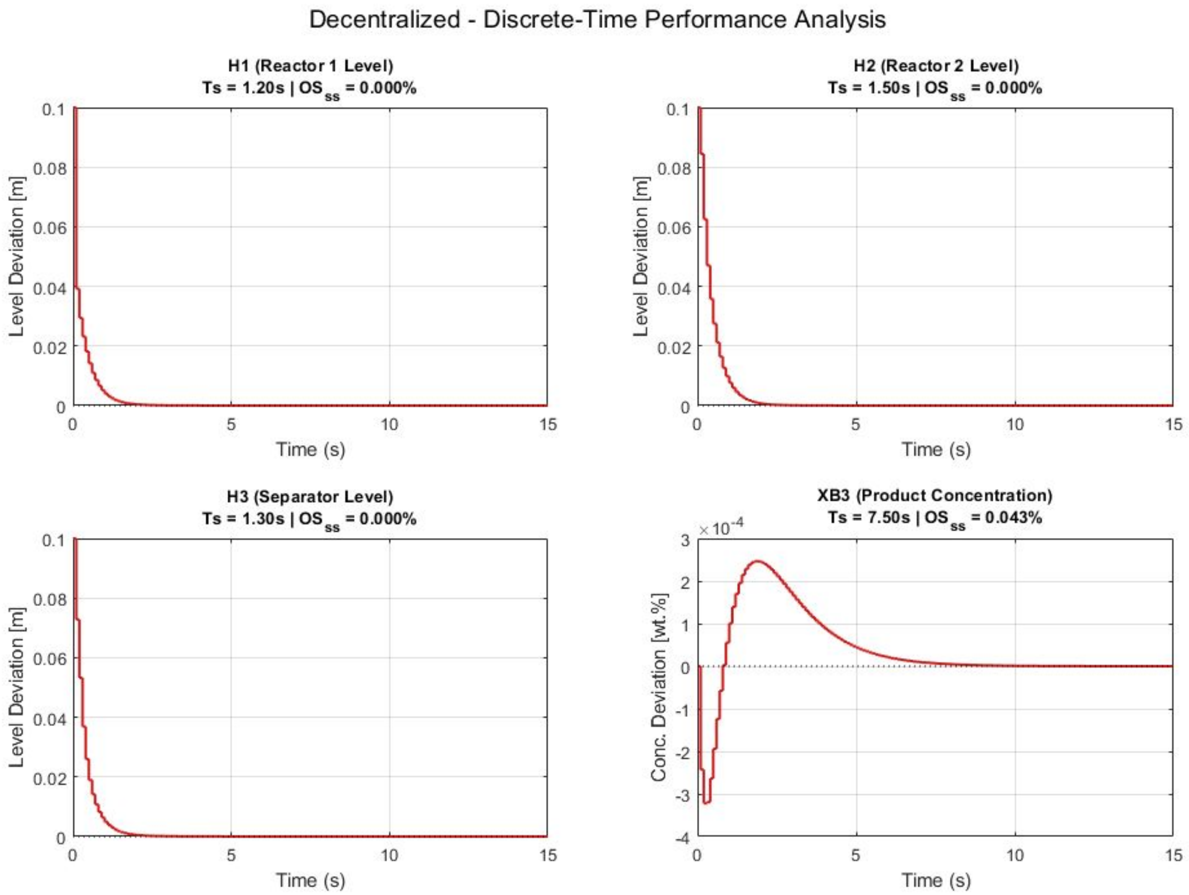


DT - CENTR - INPUT ELLIPSE - H2 MINIMIZATION – (pg.2)

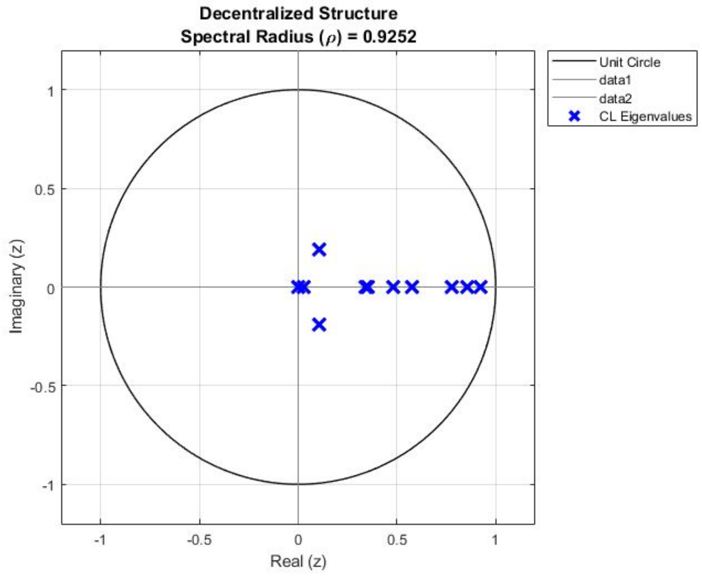




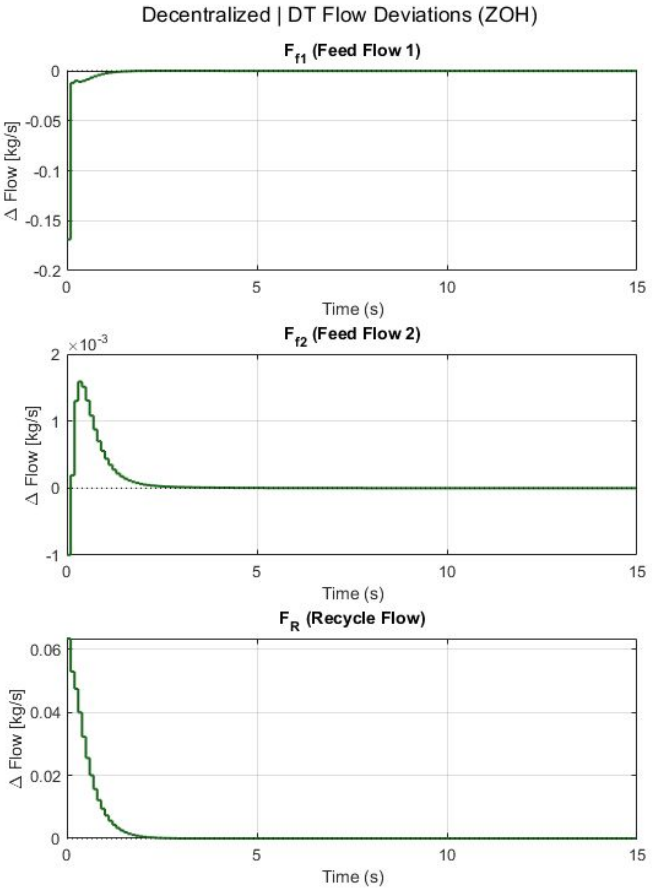
DT - DECENTR - INPUT ELLIPSE - H2 MINIMIZATION – (pg.1)

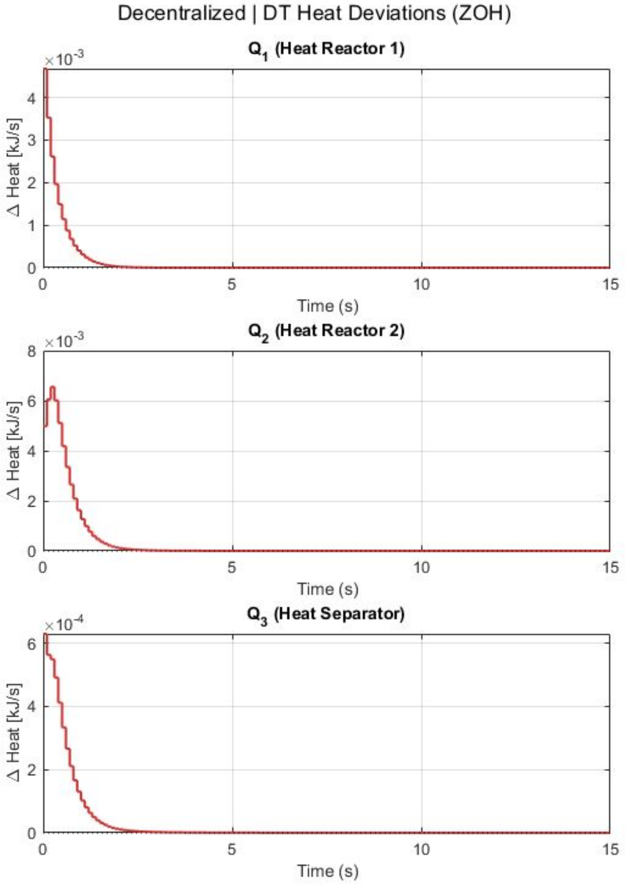


- Theoretically still controllable, but problem becomes “ill-conditioned”.
- $||K|| = 3.79$
- Worse settling times.

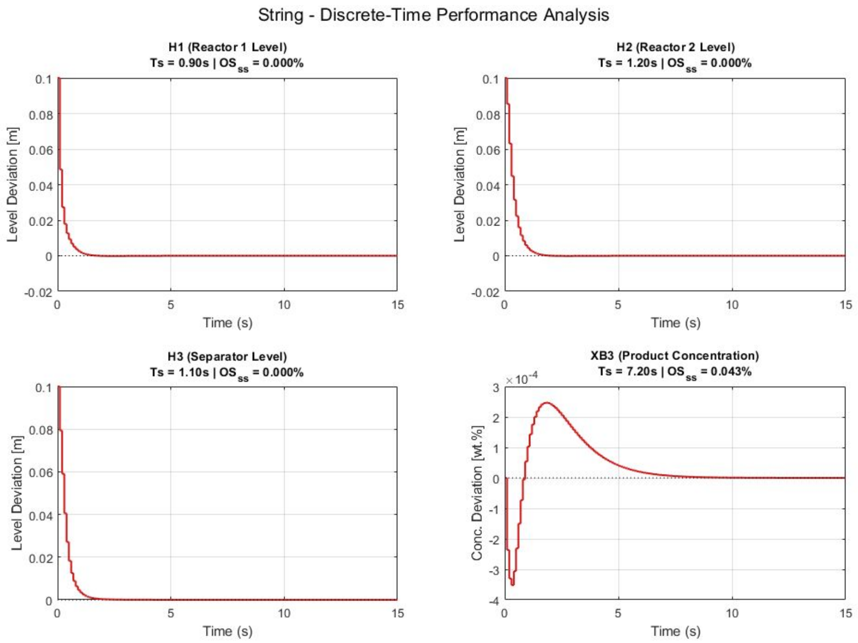


DT - DECENTR - INPUT ELLIPSE - H2 MINIMIZATION – (pg.2)





DT - STRING - INPUT ELLIPSE - H2 MINIMIZATION



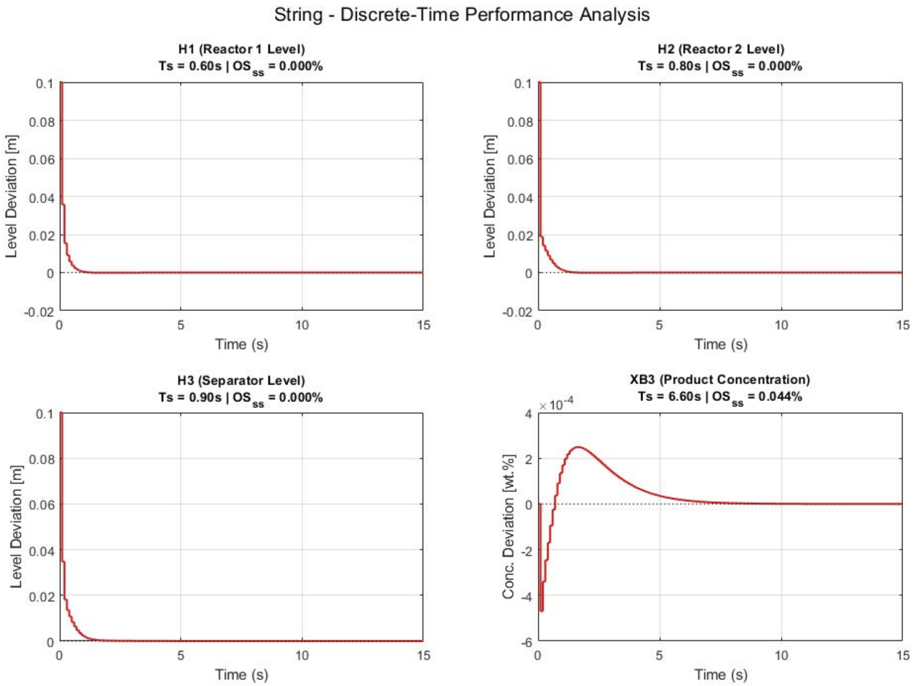
- Still ill-conditioned like the decentralized case.
- ||K|| = 4.32
- Worse settling times than the centralized case.

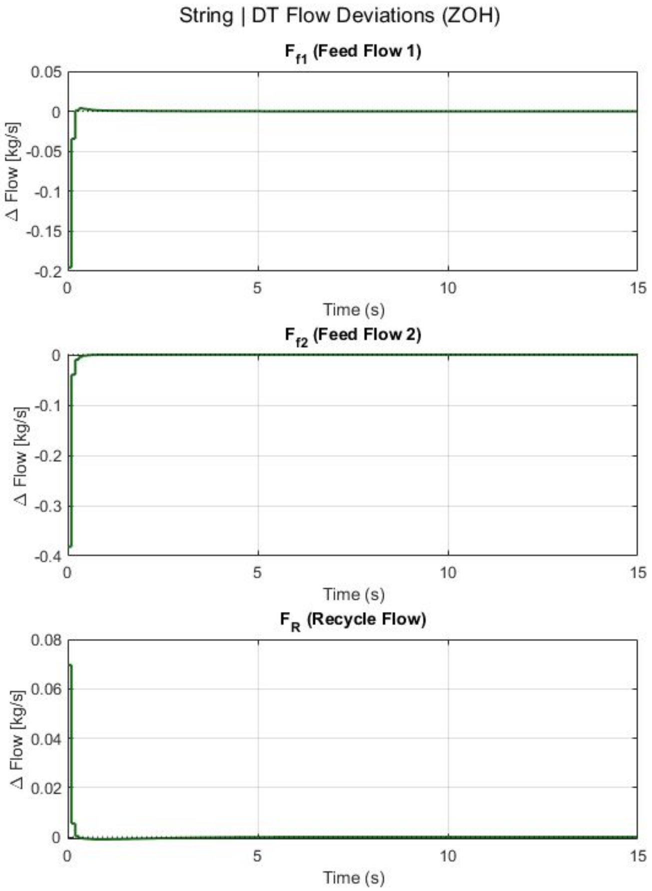
- Centralized gain matrix K below shows heavier info-exchange back and forth between reactor 1 - reactor 2 that can’t be captured by the decentralized / string.

K =

-1.0391	0.1031	0.5881	0.0148	-0.7828	0.1004	1.3068	0.0084	-0.0706	0.0020	0.0141	-0.0003
0.0947	0.0084	0.0076	-0.0066	0.0883	0.0064	-0.0025	-0.0045	0.0129	0.0002	0.0001	-0.0001
-0.2224	0.0230	-0.2091	0.0055	-0.1304	0.0171	-0.1708	-0.0044	0.0009	0.0008	-0.0056	-0.0001
0.0593	0.0087	0.0105	-0.0059	0.0825	0.0033	-0.0252	-0.0078	0.0193	0.0002	0.0003	-0.0001
-1.2980	0.2965	-4.0205	-0.0317	0.5066	0.1078	-2.8775	-0.0245	0.0702	0.0117	-0.1102	-0.0001
0.0267	0.0057	-0.0055	-0.0046	0.0480	0.0021	-0.0108	-0.0057	0.0131	0.0001	-0.0002	-0.0001

DT - STRING - ONLY H2 MINIMIZATION





Ff2 critically approaches the lower bound.

- When we drop the input constraints and employ H2 minimization alone with the same parameters, we get a “feasible” solution.
- ||K|| = 4.97
- Still worse settling times than the centralized case, but offers improvement in state trajectories over the decentralized case.

CONCLUSION

- **Like in the C.T. case, it's difficult to observe direct consequences of a global constraint on a single comparison metric or on a few states of interest.**
- **xB3 offers the main window of improvement via control strategies. For the centralised case, the shifted disk constraint provides a good trade-off, reducing xB3 settling time without introducing overshoots in reactor levels.**
- **Decentralized cases significantly tighten the achievable performance constraints. This is mainly due to strong coupling: direct flow between adjacent tanks and feedback flow from the last tank to the first.**
- **The string topology introduces a significant improvement over the decentralized case, as expected, and enables feasibility under different control criteria.**
- **H_2 control allows fast settling without overshoot in levels; however, it provides limited improvement in xB3 product concentration dynamics.**